

THE NITROGEN CURVE

Commercial Finance & Structural Margin Analysis

Institutional Equity Research | Industry Report:
Urea & Ammonia

Research Scope & Modeling Disclaimer

Educational Portfolio Analysis: This document has been prepared solely as an educational and portfolio-analysis exercise to demonstrate advanced financial modeling, operational analysis, and corporate scenario-planning methodologies.

Data Sourcing Framework: The underlying analysis is constructed utilizing publicly available corporate filings, investor presentations, industry publications, and historical market datasets. Certain logistical parameters, freight tariffs, carbon cost adjustments, and plant operating run-rates are illustrative and range-bound to protect proprietary model configurations.

Scenario Parameters & Methodology Notice: The financial schedules, multi-variable sensitivity matrices, and Through-the-Cycle (TTC) valuation outputs contained herein are intended to demonstrate flexible, scalable analytical models under varying macroeconomic environments. They are not intended to predict future spot market realizations, equity pricing movements, or definitive corporate performance outcomes.

No Investment Advice: This report does not constitute investment advice, equity research recommendations, or a solicitation to buy or sell securities. All analytical evaluations represent independent, model-driven conclusions for educational evaluation purposes.

Industry & Market Framework

Nitrogen Value Chain Overview

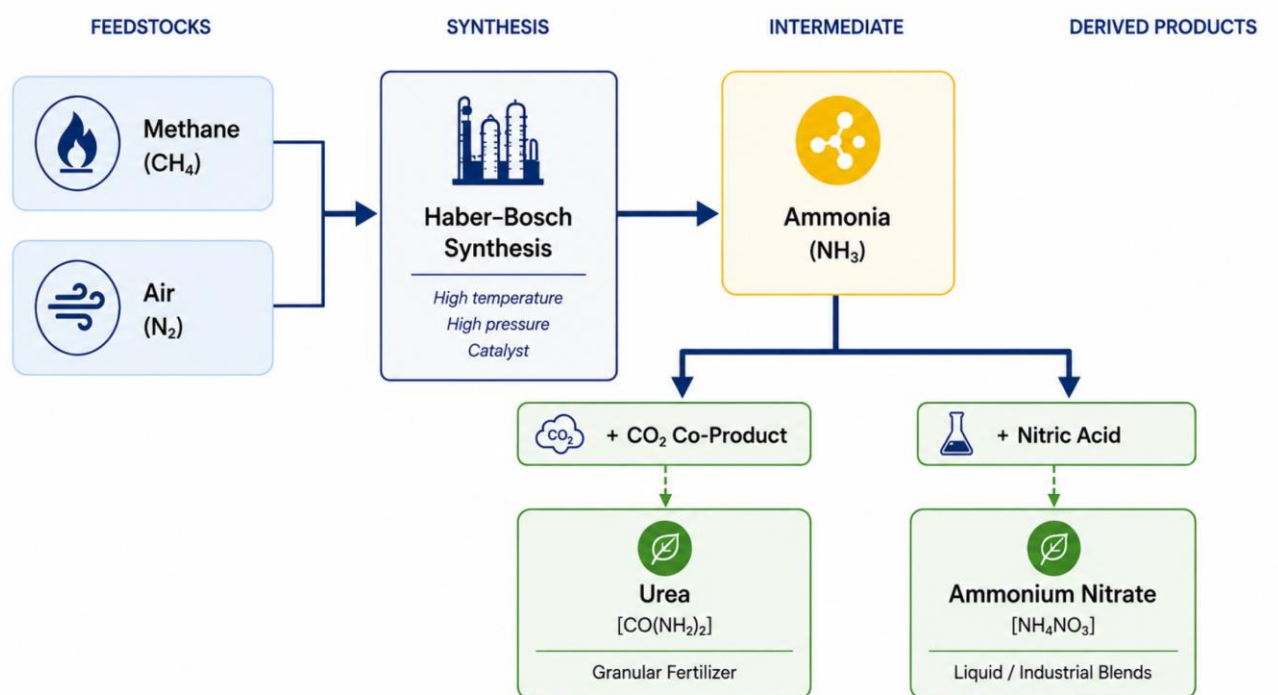
Canada's nitrogen fertilizer sector occupies a structurally advantaged position on the global cost curve, sustained by abundant low-cost natural gas feedstock, world-class production assets, and direct access to premium North American agricultural markets.

Nitrogen, phosphate, and potash form the triad of essential primary agricultural nutrients. Among nitrogen inputs, urea $\text{CO}(\text{NH}_2)_2$ remains the dominant global molecule due to its high nitrogen concentration (46%), handling cost-efficiencies, and minimal soil salinity impact relative to alternative nitrogen carriers. While agriculture accounts for over 70% of global urea consumption, industrial applications—including diesel exhaust fluids (DEF), resins, melamine, and animal feed additives—provide highly resilient, non-cyclical demand baselines.

Feedstock Synthesis & Process Flow

Industrial nitrogen production relies on a two-stage chemical synthesis where natural gas (methane) serves as both the primary thermal energy source and the hydrogen feedstock.

Nitrogen Fertilizer Value Chain (Simplified)



a) Ammonia Synthesis:

Methane is steam-reformed to extract hydrogen, which is reacted with atmospheric nitrogen via the Haber-Bosch process to yield anhydrous ammonia (NH₃). Ammonia is the highly concentrated chemical foundation of the entire chain, consisting of 82% pure nitrogen content.

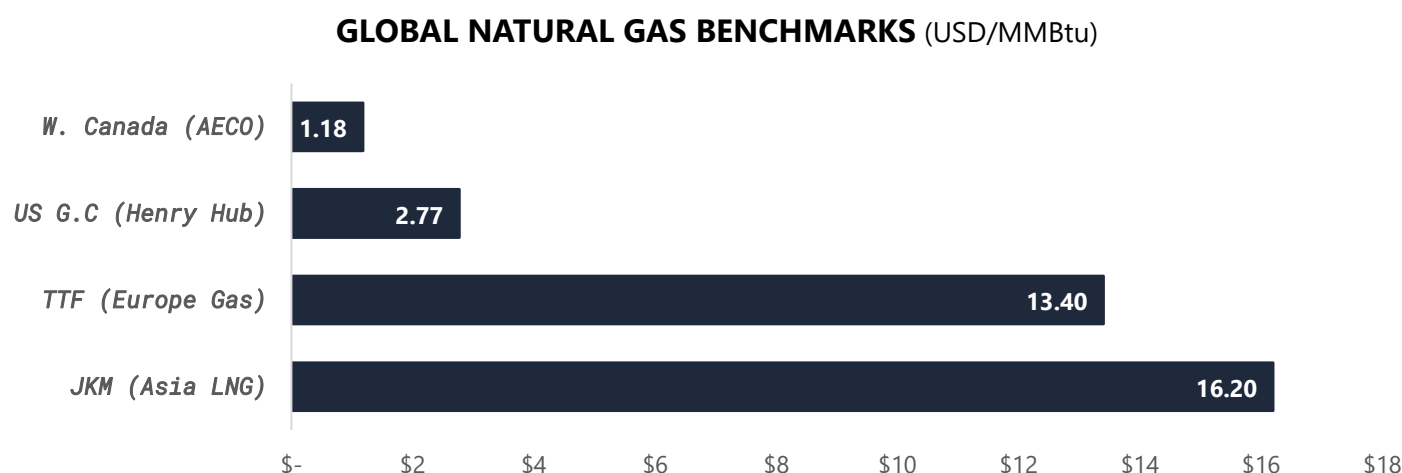
b) Downstream Upgrading:

Ammonia is either sold directly or upgraded into downstream derivatives to alter its chemical properties and nitrogen delivery:

- Granular Urea $\text{CO}(\text{NH}_2)_2$: Produced by reacting NH_3 with captured co-product carbon dioxide under high pressure. It yields a dry granular fertilizer with 46% nitrogen content.
- Ammonium Nitrate $(\text{NH}_4)\text{NO}_3$: Synthesized by reacting NH_3 with nitric acid, yielding an industrial/liquid blend featuring 29% to 35% nitrogen content.
- Urea Ammonium Nitrate (UAN): A liquid upgraded fertilizer created by combining liquid urea and ammonium nitrate solutions with water. It carries a final concentration of 28% to 32% nitrogen content, depending on the customer's technical specification.

Feedstock Economics & Cost Curve Positioning

- Because natural gas represents 70% to 80% of the cash cost of ammonia/Urea production, spread firmly anchors Canadian facilities in the bottom decile (lowest cost) of the global fertilizer cost curve. Western Canadian producers benefit from a systemic geographic discount; local AECO natural gas hub pricing historically trades at a deep discount to global LNG-linked benchmarks, notably European Title Transfer Facility (TTF) and Asian Japan Korea Marker (JKM) prices which insulates local operators from global energy shocks impacting overseas competitors.



➤ **Structural Margin Cushion:**

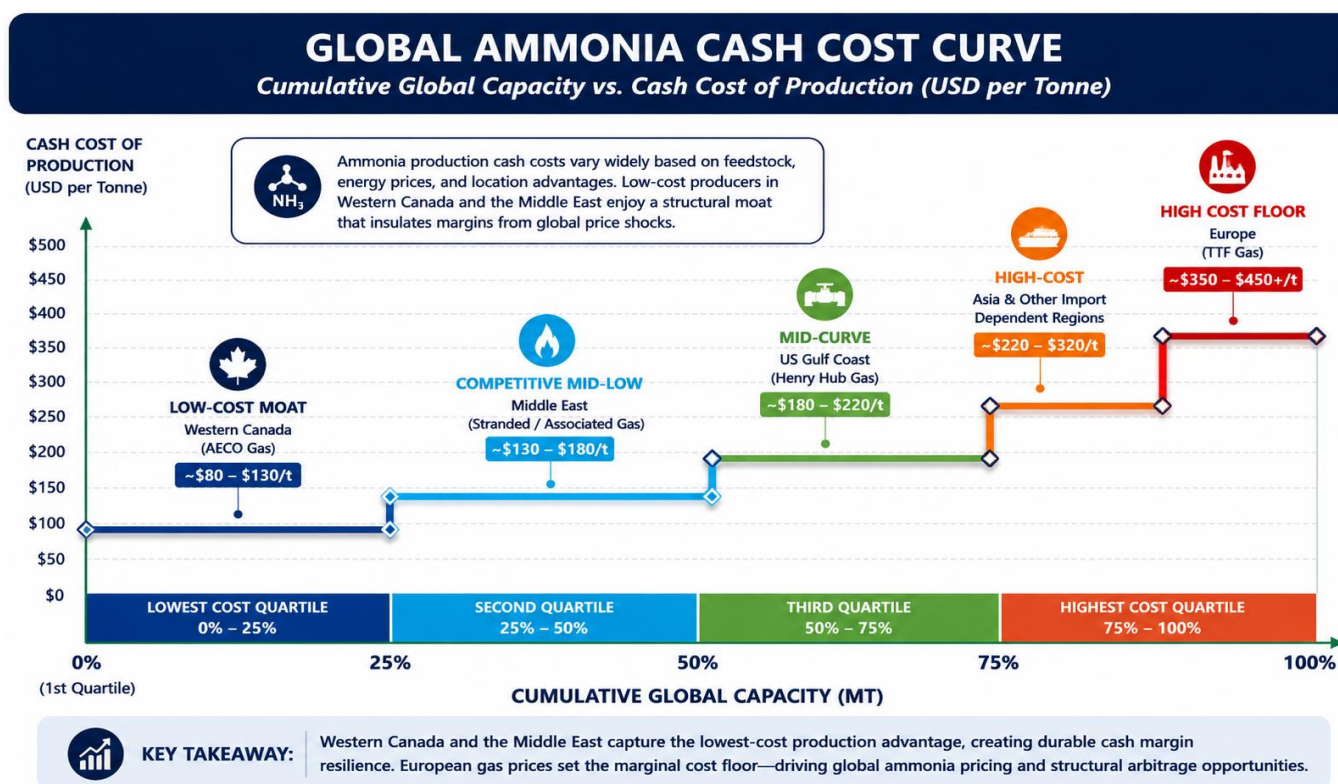
Under the assumptions used in this report, AECO pricing remains materially below Henry Hub and international gas benchmarks against US peers (Henry Hub) and a more than \$12.00/MMBtu structural advantage against European/Asian marginal producers.

Global Nitrogen Cash Cost Curve

Feedstock Economics Drive Structural Margin Advantage

Natural gas represents the primary variable input cost in industrial ammonia synthesis and downstream nitrogen fertilizer manufacturing. As a result, regional gas benchmarks create a highly differentiated global manufacturing cost structure, establishing a clear multi-tiered hierarchy of cash production costs across global production hubs.

Exhibit 1: Upstream Cost Insulation (Ammonia Base-Level Production)



The approximately three-to-five-fold spread between first-quartile and marginal-cost ammonia production regions demonstrates how feedstock economics and geography materially alter nitrogen manufacturing competitiveness. Producers with access to stranded or discounted gas supplies enjoy a durable structural advantage independent of fertilizer market cycles.

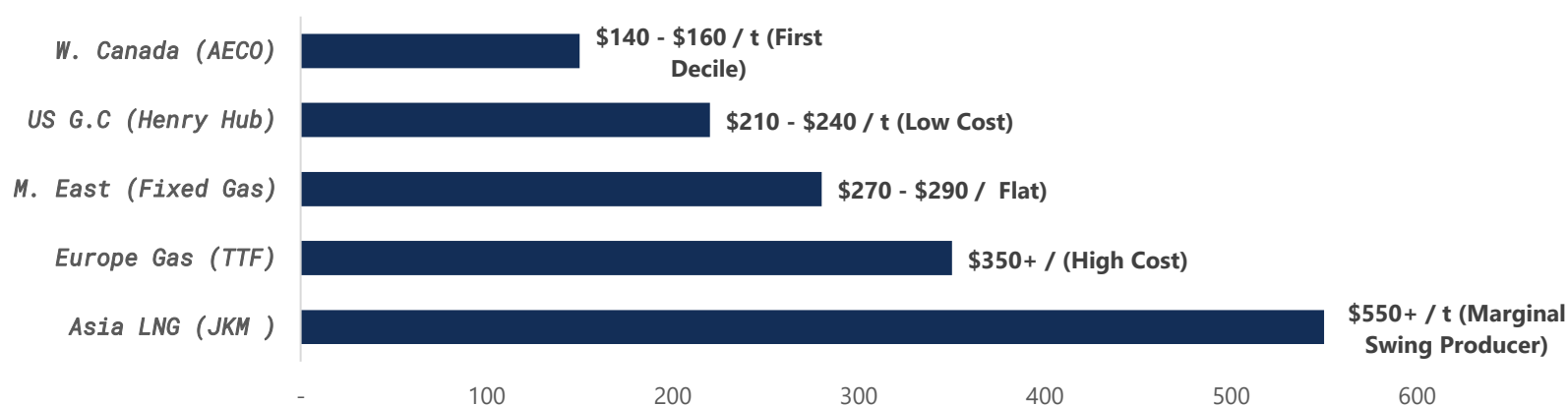
This protective footprint insulates upstream corporate margins long before final upgrading or localized distribution networks take effect.

Global Nitrogen Production Cost Benchmarking

Natural gas represents the primary variable input cost in ammonia and nitrogen fertilizer production. As a result, regional gas benchmarks create a highly differentiated global manufacturing cost structure, positioning producers at distinct locations along the nitrogen cash cost curve.

- **Exhibit: Global Nitrogen Cash Cost Curve**

GLOBAL NITROGEN CASH COST CURVE BY PRODUCTION HUB



Western Canada (AECO) -- First Plateau — Absolute Cost Leadership

- Feedstock: Sub-\$1.20/MMBtu AECO gas
- Estimated Cash Cost: ~\$140–160/t
- Position: Lowest-cost quartile globally

Western Canadian producers occupy the absolute bottom of the global cost curve. Access to deeply discounted, landlocked gas provides substantial insulation from international energy volatility, supporting positive free cash flow generation even during severe nitrogen downturns.

Low-Cost Tier — US Gulf Coast (Henry Hub)

- Feedstock: Henry Hub Natural Gas
- Estimated Cash Cost: ~\$210–240/t
- Position: Low-Cost Global Producer

US Gulf Coast producers leverage abundant, highly flexible domestic gas infrastructure to secure premier North American positioning. However, because Henry Hub gas pricing matches broader market-clearing levels, these facilities operate at a premium relative to Western Canadian fields.

Mid-Curve Flat Tier — Middle East (Fixed Gas)

- Feedstock: Government-regulated gas contracts
- Estimated Cash Cost: ~\$270–290/t
- Position: Mid-Curve Stable Cost Producer

While exceptionally low-cost during primary ammonia synthesis, Middle Eastern producers benefit from long-term regulated gas contracts that support stable cost structures, positioning them in the mid-cost segment of the global nitrogen industry. They remain highly stable and predictable but trade away absolute cost leadership to top-tier North American hubs.

High-Cost Tier — Europe Gas (TTF)

- Feedstock: LNG-Linked TTF Gas Pricing
- Estimated Cash Cost: ~\$350-450+/t
- Position: High-Cost International Producer

European nitrogen operations face constant margin erosion due to reliance on expensive imported gas frameworks. Prolonged exposure to TTF pricing routinely triggers industrial production curtailments during regional energy market shocks.

Marginal Swing Producer — Asia LNG (JKM)

- Feedstock: Imported LNG
- Estimated Cash Cost: ~\$550+/t
- Position: Marginal Swing Producer

High-cost manufacturers dependent on JKM-linked Asian LNG sit at the extreme end of the cost curve. These facilities represent the highest-cost supply segment and establish the international clearing price floor for globally traded nitrogen derivatives during tight market balances.

Investment Implications

The approximately \$400/t spread between Western Canadian producers and Asian LNG-based producers highlights the magnitude of the industry's structural cost advantage.

This positioning provides:

- First-decile cost leadership
- Enhanced free cash flow resilience
- Lower break-even fertilizer pricing
- Reduced exposure to global gas-price shocks
- Superior through-cycle margin durability

Key Takeaway

Cost-curve position is structural, not cyclical. Western Canadian nitrogen assets operate within the first decile, supported by access to discounted AECO feedstock. This structural advantage underpins profitability resilience and creates a durable competitive moat relative to higher-cost global producers.

Global Macro Drivers & Affordability

❑ The Affordability Dynamic & Crop Economics Linkage

Global nitrogen and potash demand frameworks are strictly tied to farm-level economics, primarily driven by corn and wheat price behaviors. When crop prices rise, grower margins expand, maximizing fertilizer application rates to optimize crop yields.

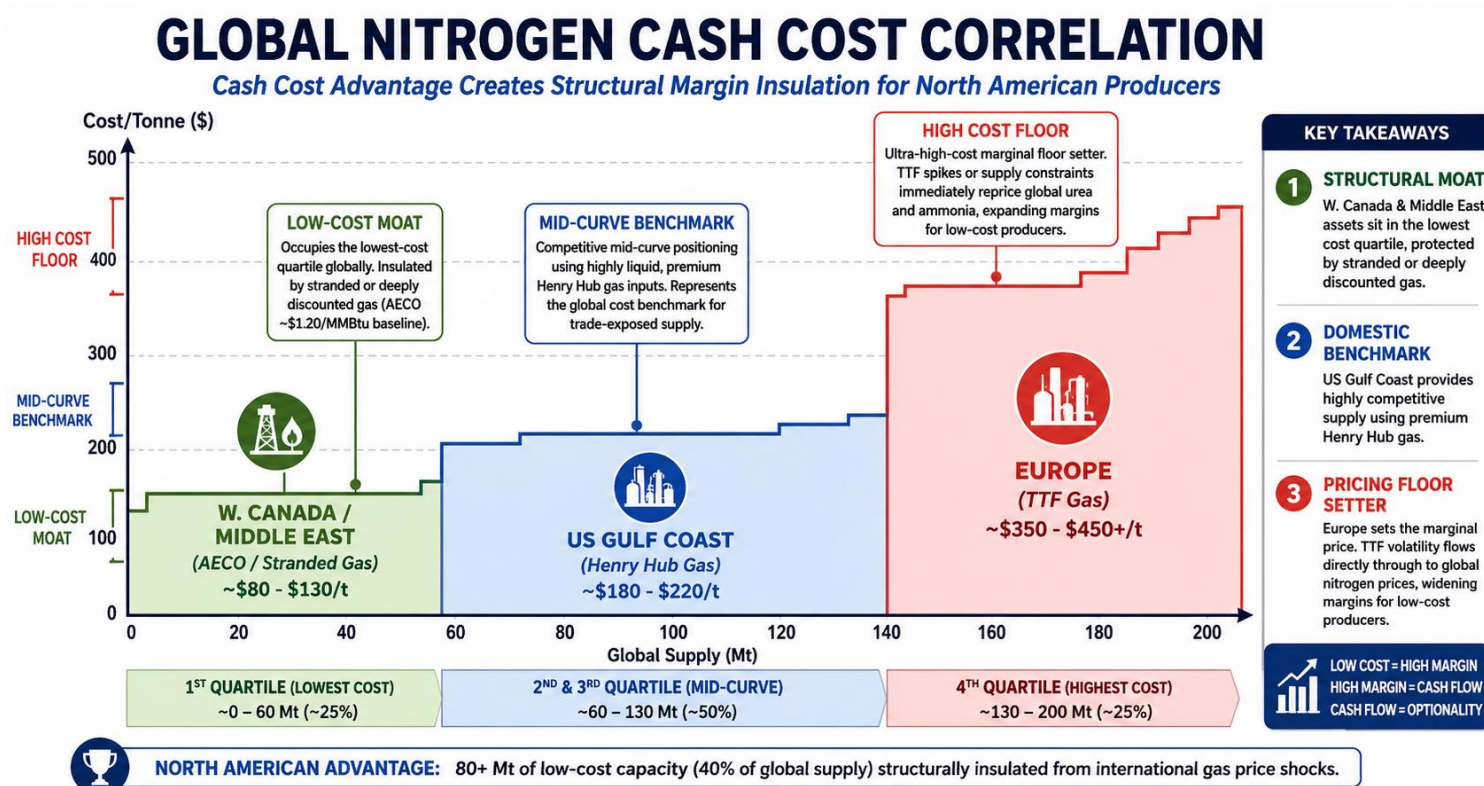
The primary metric tracking this relationship is the **Nutrient Affordability Index** (the ratio of fertilizer prices to crop prices). The forecast assumes a stable normalization where ratios compress toward a historical **0.15 raw ratio baseline**, signaling robust direct-to-grower retail demand and supporting stable sales volume schedules across the 2025F–2026F period.

❑ Global Supply/Demand Dynamics

Global Nitrogen Demand Growth YoY: Global nitrogen demand continues to exhibit a highly resilient **1.5% to 2.0% structural year-over-year growth trend line**, fueled by global population expansions, dietary shifts toward higher protein consumption, and intensifying industrial requirements (such as Diesel Exhaust Fluid and industrial resins).

❑ Global Nitrogen Cost Curve & Competitive Positioning

The global nitrogen floor price is fundamentally set by the highest-cost, marginal producers on the global cost curve—historically European manufacturers reliant on volatile, imported liquefied natural gas (LNG) exposed to European TTF pricing benchmarks.

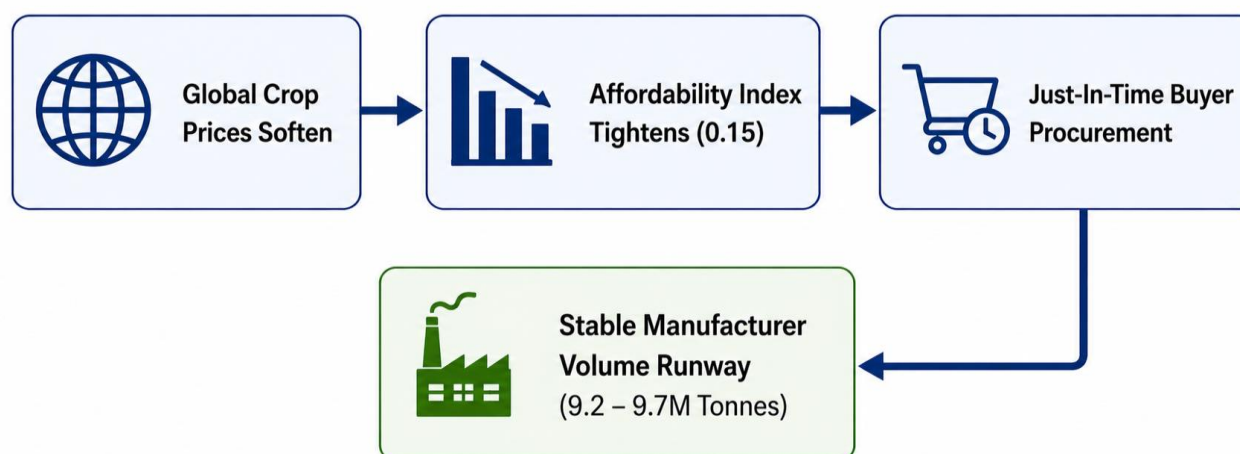


Crop Economics & Affordability Analysis

Global nitrogen demand functions as a direct derivative of agricultural net cash income, exhibiting high cyclical sensitivity to the global grain complex. Coarse grains such as corn and wheat require heavy nitrogen applications to optimize yield profiles. Consequently, fertilizer procurement schedules are tightly bound to farm-level affordability dynamics.

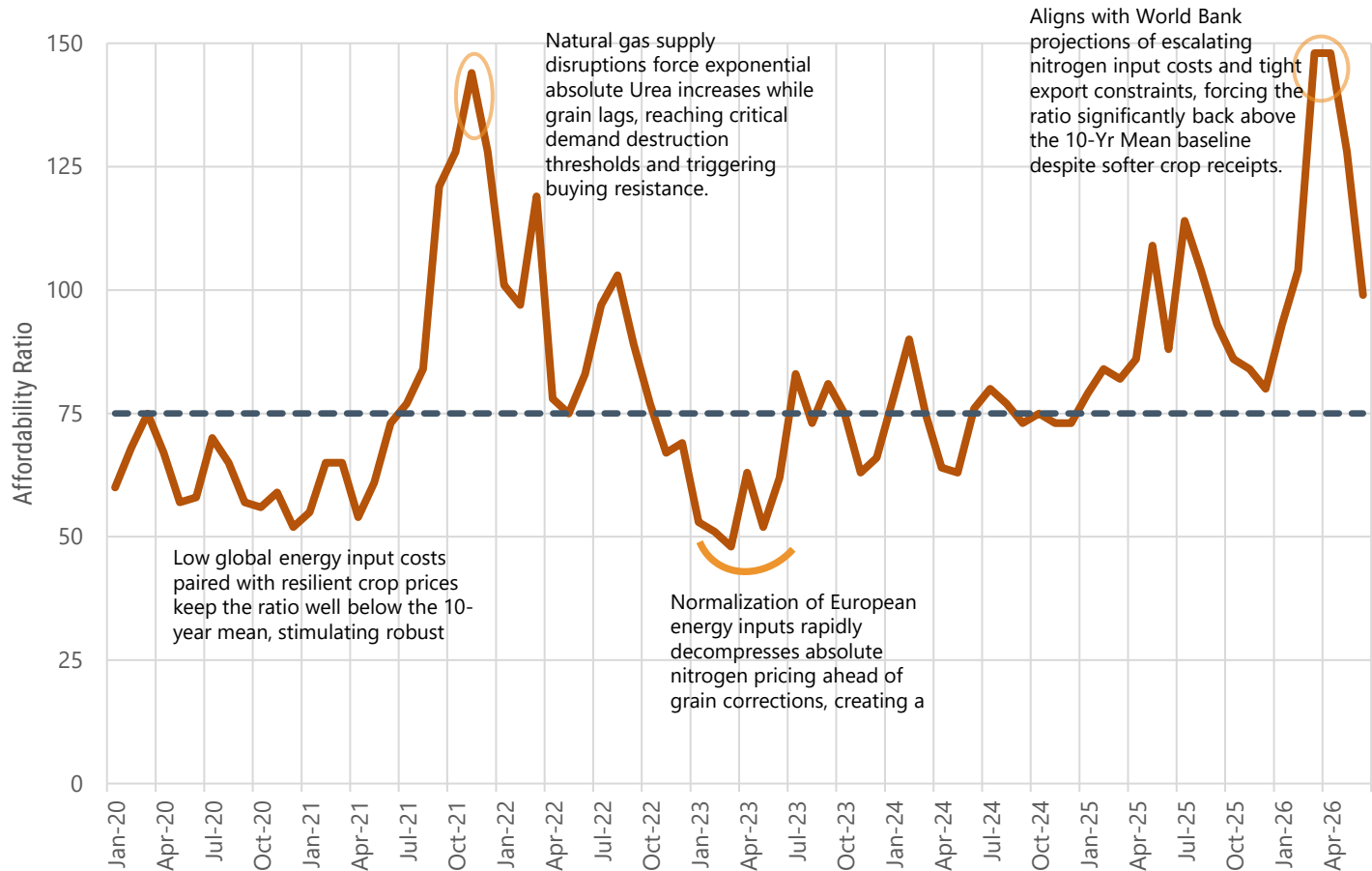
To evaluate forward demand pathways, institutional research relies on the **Urea-to-Corn Affordability Index** (tracked by Bloomberg Intelligence / Green Markets and Farmers Business Network). This index calculates the nominal price of granular urea relative to front-month grain futures.

- ❑ **The Affordability Dynamic:** Data indicates that while absolute global urea prices have stabilized below their historic 2022 supply-shock peaks, a parallel softening in global crop receipts has kept the relative affordability index compressed near 0.15 points. According to FBN research, this requires a historically higher number of grain bushels to cover baseline nutrient costs, creating localized demand resistance and compressed buying windows.
- ❑ **World Bank Macro Overlay:** According to the World Bank Commodity Markets Outlook, the global fertilizer price index is projected to rise by more than 30% across 2026. This trajectory is fundamentally supported by escalating input costs—particularly for nitrogen- and phosphate-based configurations—juxtaposed against highly resilient global structural demand. Prices are mathematically modeled to ease into 2027 as localized export constraints untangle and secondary production capacities come online.
- ❑ **Demand Volume Runway:** Despite near-term margin pressure on farmers, baseline volume demand remains structurally protected by global food security requirements. Leading global producer guidance (Nutrien Ltd.) projects resilient core utilization, modeling forward North American nitrogen volumes within a disciplined 9.2 to 9.7 million tonne baseline. This confirms that while producer pricing power faces cyclical adjustments, physical consumption volumes remain anchored.



Data indicates that while absolute global urea prices have stabilized below their historic 2022 supply-shock peaks, a parallel softening in global crop receipts has kept the relative affordability index compressed near a raw ratio of 0.15. According to FBN research, this requires a historically higher number of grain bushels to cover baseline nutrient costs, creating localized demand resistance and compressed buying windows.

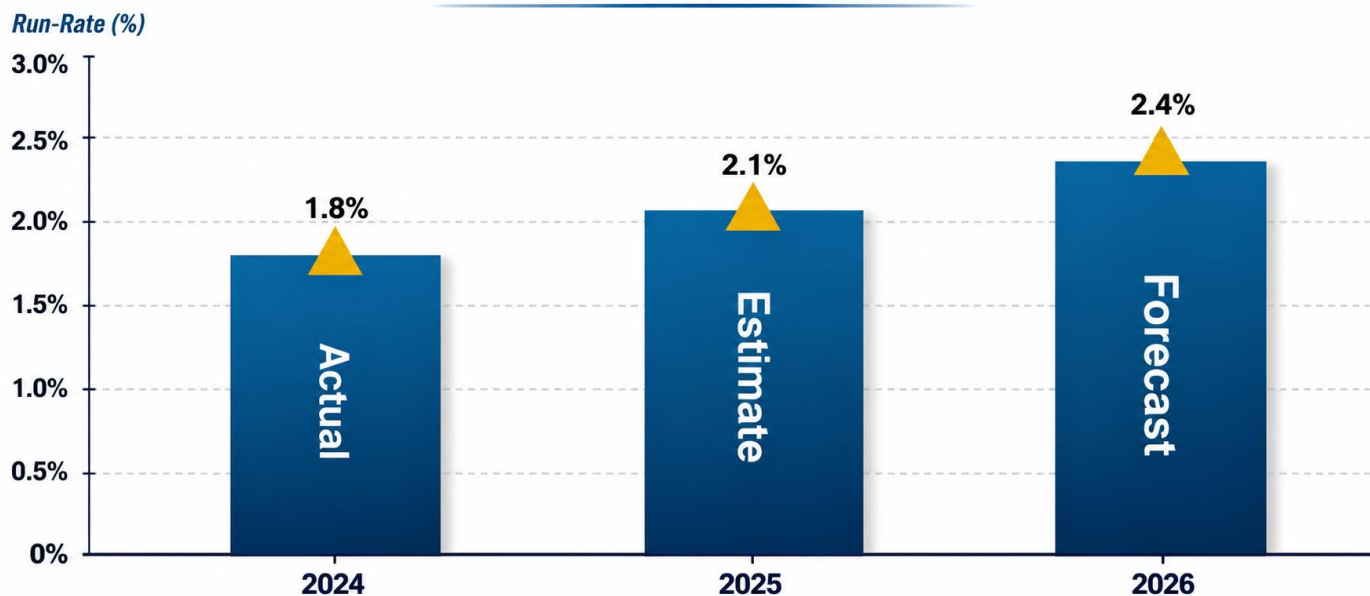
CROP-NUTRIENT AFFORDABILITY CORRELATION



Global Supply/Demand Balances & Cost Curve Positioning

On a macro level, global nitrogen and potash consumption continue an upward multi-year trajectory, driven by intensifying yield demands per arable hectare.

GLOBAL NITROGEN DEMAND RUN-RATE & FORECAST



- **Nitrogen Demand Run-Rate:** According to data from The World Bank Commodity Markets Outlook (April 2026 Baseline) and Nutrien Ltd. Market Outlook (April 2026), global nitrogen demand growth accelerated from 1.8% in 2024 to an estimated 2.1% in 2025. As displayed in the chart, industry forecasts generally indicate growth above long-term historical averages for 2026, outpacing historical 10-year trend lines of 1.5%. This growth profile is fundamentally propelled by aggressive grain cultivation expansions in Latin America and urgent domestic food security consumption mandates in South Asia.
- **Potash Demand Run-Rate:** In tandem, drawing from the same World Bank and Nutrien industry baselines, global potash shipments are projected to stabilize at record bounds between 74 to 77 million tonnes in 2026. This volume plateau reflects major structural crop application catch-up across key offshore regions, following multiple cycles of severe Eastern European export and logistical supply curtailments.

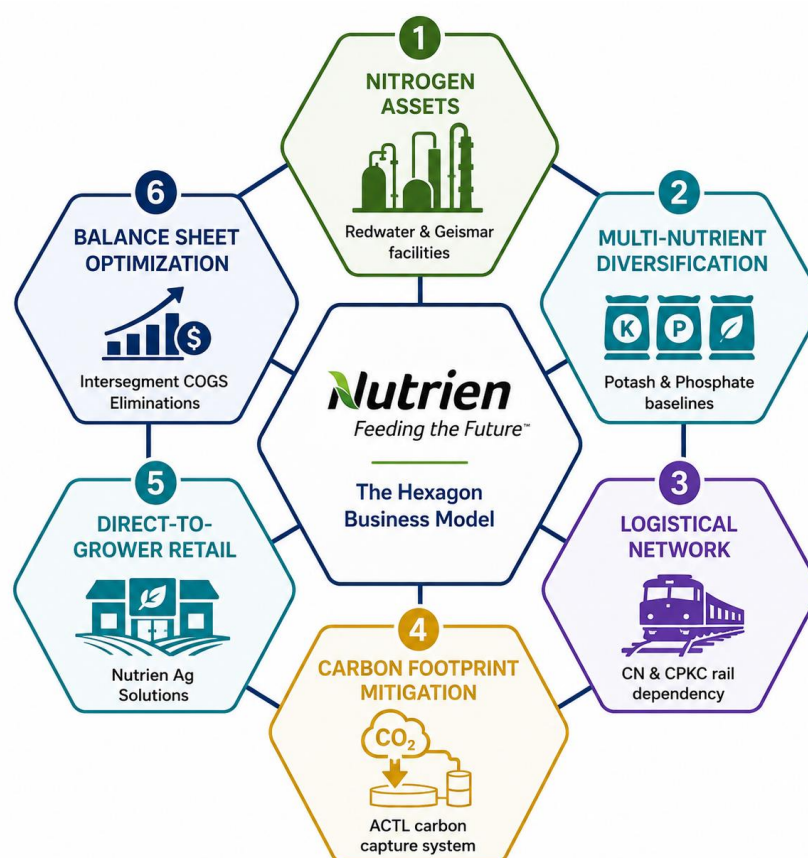
Regulatory & Decarbonization Considerations

While highly profitable, nitrogen synthesis is inherently carbon-intensive, exposing producers to evolving environmental regulatory overlays. To mitigate compliance costs and capture secular premiums for low-carbon industrial inputs, the sector is aggressively allocating capital to Carbon Capture, Utilization, and Storage (CCUS) and "Blue Ammonia" infrastructure (ammonia produced via natural gas where the co-product CO₂ is sequestered).

Regulatory Note: Carbon pricing frameworks continue to evolve and remain a key modeling variable within long-term valuation assumptions.

Business & Operating Model Assessment

Nutrien Business Overview



Nutrien operates a top-tier global nitrogen asset base, currently ranking as the third-largest nitrogen producer globally and the second-largest in North America by volume, commanding approximately three percent of global nitrogen nameplate capacity. The company's upstream network encompasses an annual gross ammonia nameplate capacity of 7.3 million tonnes across its core facilities. In its latest full-year operational disclosures, Nutrien recorded total upstream sales of 27.5 million tonnes of essential fertilizer products across all nutrient segments. Within the nitrogen segment specifically, the company generated annual sales volumes of approximately 10.6 million tonnes of combined ammonia, urea, and upgraded solutions. Backed by extensive reliability optimization initiatives and targeted debottlenecking programs across its core North American facilities, Nutrien operated at an advanced institutional utilization rate of 89% to 93% of gross nameplate capacity.

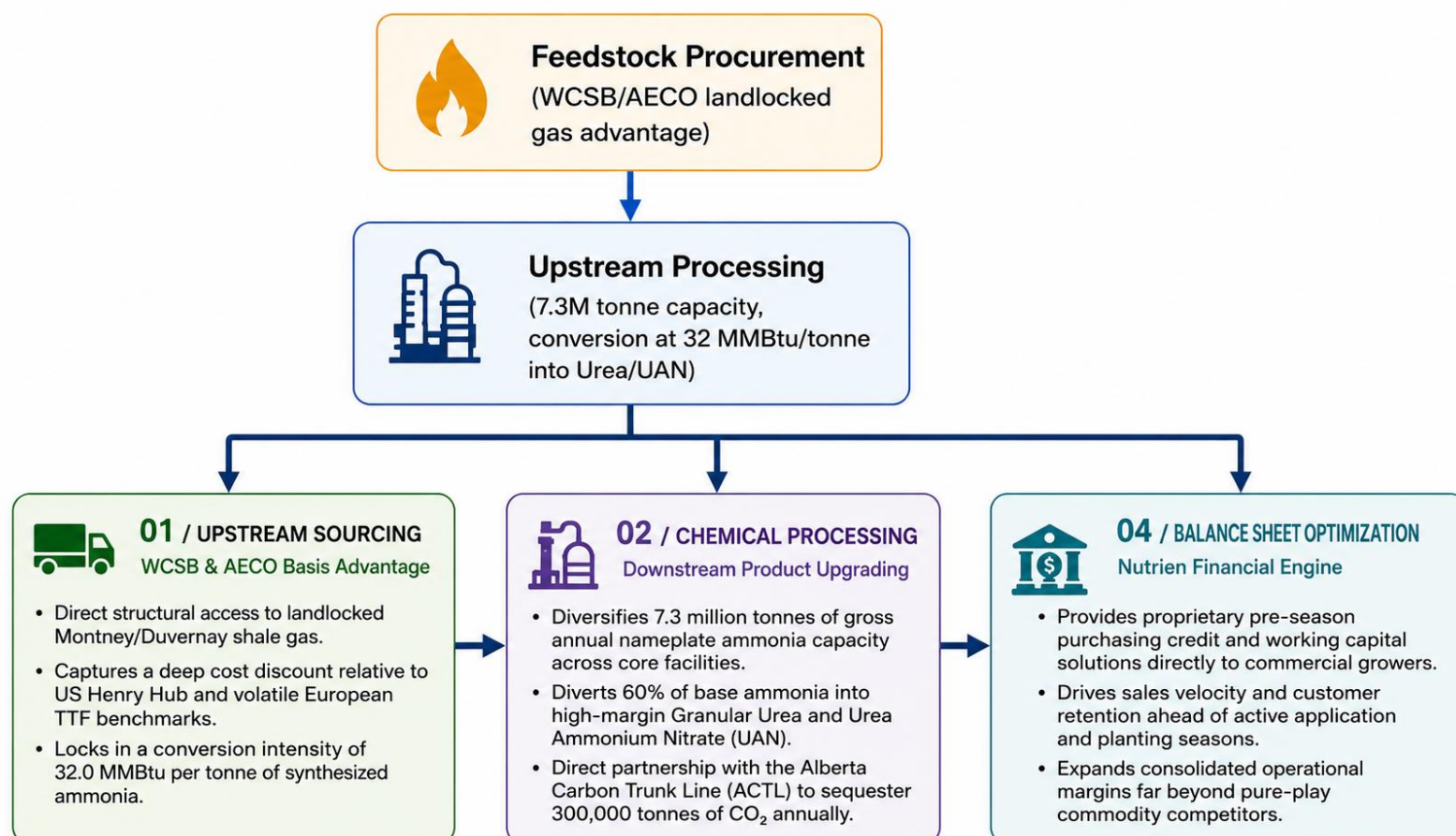
Based on recent public disclosures of capital allocation priorities aimed at maximizing near-term free cash flow yield, corporate development has adjusted its upstream growth pipeline. In line with a capital expenditure guidance ceiling of \$2.0 billion to \$2.1 billion, Nutrien intentionally deferred its massive greenfield clean ammonia export projects to focus capital strictly on low-risk, high-return brownfield optimizations. The primary active operational expansion centers on the sequential low-cost debottlenecking and carbon-capture optimization projects at its Redwater, Alberta and Geismar, Louisiana facilities. The Redwater carbon system expansion, engineered in partnership with the Alberta Carbon Trunk Line (ACTL), is fully online and capable of sequestering up to 300,000 tonnes of CO_2 annually. This carbon footprint mitigation is highly strategic, directly insulating Nutrien's Western Canadian cash margins against Canada's revised industrial carbon pollution pricing trajectory, which is legally capped at a maximum of \$130 per tonne.

The underlying structural unit economics of the nitrogen segment rely heavily on a highly localized feedstock advantage. Nutrien's Western Canadian assets draw natural gas primarily from the Montney and Duvernay shale formations within the Western Canadian Sedimentary Basin (WCSB), sourced directly from major regional pipeline operators. From an operational efficiency standpoint, the manufacturing framework requires an intensive natural gas consumption multiplier of 32.0 MMBtu per single tonne of ammonia produced.

Because these assets are positioned adjacent to the inland AECO pricing hub—which is structurally discounted relative to the North American Henry Hub benchmark and trades at a fraction of volatile European TTF prices—Nutrien maintains a permanent, highly insulated margin cushion that protects corporate EBITDA from cyclical commodity downturns. Downstream processing workflows dictate that approximately 60% of Nutrien’s gross ammonia production is diverted and upgraded into high-margin nitrogen solutions, primarily downstream solid Urea and Urea Ammonium Nitrate (UAN). The remaining 40% of un-upgraded ammonia is either commercialized directly as anhydrous ammonia or routed to supplement chemical inputs in neighboring operations, including the company's localized phosphate processing facilities.

Nutrien’s outbound distribution footprint splits its market delivery between highly profitable domestic channels and volatile global export trade. Of its total manufactured upstream fertilizer volumes, approximately half is allocated to the international export market via specialized deep-water gateways, with major volumes flowing to high-demand agricultural regions in South America (principally Brazil) and Asia. The remaining approximately half of production is retained within the local North American corn and wheat belts. This domestic volume is distributed directly through Nutrien Ag Solutions' massive proprietary downstream retail network, which comprises over 2,000 retail locations globally and commands an extensive, vertically integrated footprint across rural Canada and the United States.

To maximize value capture across this unique, direct-to-grower retail footprint, Nutrien leverages Nutrien Financial—a dedicated, captive financial services division. Operating primarily across its high-volume US and Australian retail branches, Nutrien Financial provides tailored credit solutions, working capital financing, and flexible pre-season purchasing programs directly to commercial farmers. By vertically integrating credit financing into product sales, Nutrien significantly drives sales velocity for its high-margin proprietary crop nutrients and protection lines, locks in customer retention ahead of active planting seasons, and secures steady non-commodity interest revenue. This proprietary financial engine ultimately acts as a powerful balance sheet optimizer, lowering credit default risk through localized underwriting while expanding the company’s aggregate retail operating margins far beyond those of pure-play commodity competitors.



Reconciling Cost Distribution to Consolidated Outflows

The Nutrien Functional COGS visual demonstrates that downstream distribution channels anchor Nutrien's consolidated spending framework. On a gross pre-elimination basis, **Retail Inventory & Procurement commands the single largest share at 53.3%** of the corporate expense pie. When fixed overhead (18.0%), commodity mining/production costs (18.8%), logistics freight (4.2%), and corporate expenses (3.9%) are layered on, they establish the total gross operational cost pool before internal corporate netting.

To reconcile this gross visual with the final Income Statement projections, the model processes **\$1.39 billion in Intersegment COGS Eliminations** and **\$10.56 billion in Retail COGS Eliminations**, which reflects product manufactured upstream and transferred internally to be sold through Nutrien Ag Solutions retail branches.

Once these adjustments are finalized and the **\$4.92 billion Operating Absorption Delta** is factored in, the net operational outflow yields a **Total Cost of Goods Sold of \$17.61 billion for 2025F and \$17.64 billion for 2026F**. This structural breakdown confirms that while Nutrien captures massive, low-cost margin spreads on its upstream nitrogen assets, the consolidated financial statement is driven heavily by a stable, low-volatility retail distribution model.

CF Industries Business Overview

CF Industries operates as the world's largest corporate producer of ammonia on a gross basis, commanding an average annual capacity of 10.5 million short tons of gross ammonia across its production footprint. Unlike Nutrien's diversified agronomic and multi-nutrient profile, CF functions as a pure-play nitrogen giant, utilizing its production assets to manufacture upgraded derivatives including granular urea, urea ammonium nitrate (UAN), and industrial ammonium nitrate (AN). CF routinely operates at an industry-leading capacity utilization rate of 95% to 97%, outpacing its North American peer averages by nearly 10% due to aggressive reliability investments and centralized asset design.

The crown jewel of CF's international manufacturing footprint is its Donaldsonville, Louisiana complex—the largest nitrogen facility in the world. However, its primary competitive cost moat lies in its Western Canadian asset base: the Medicine Hat Complex in Alberta. As Canada's largest operating nitrogen manufacturing complex, Medicine Hat features an annual marketable capacity of approximately 1.25 million tons of gross ammonia and 810,000 tons of granular urea.

Sourcing Moats & Regulatory Tax Exposure

Similar to Nutrien's regional footprint, the Medicine Hat asset utilizes its direct proximity to landlocked Western Canadian Sedimentary Basin (WCSB) gas, securing natural gas inputs at a deep, systemic discount via the AECO pricing hub relative to Henry Hub and European TTF benchmarks. From a regulatory standpoint, this pure-play profile heightens CF's vulnerability to shifting carbon tax regimes compared to diversified peers. To mitigate Canada's output-based carbon pricing systems, CF is executing targeted carbon-capture and underground sequestration retrofits, laying the groundwork for a transition toward low-carbon "blue" ammonia production. This strategy allows the company to shield its high-margin Canadian assets from severe compliance penalties while positioning itself to capture lucrative export premiums in the emerging industrial clean energy and marine fuel sectors.

Upstream Unit Economics & Consolidated Cost Anatomy

A granular analysis of our 2026F Forecast Cost Schedule reveals a distinct operational paradox within Nutrien's financial architecture: Consolidated corporate margins are highly insulated from domestic natural gas price spikes, yet remain completely unanchored on the global margin expansion curve.

For the 2026 forecast period, Nutrien's total consolidated Cost of Goods Sold (COGS) is projected at \$17.64 billion. A structural breakdown of this outflow shows that downstream asset classes—specifically the Downstream Retail Segment COGS at \$13.23 billion (occupying 74.2% of unadjusted cash costs) and Upstream Fixed Overhead at \$4.47 billion—overwhelmingly dominate corporate spending.

Within the upstream Nitrogen production stack (\$10.06 million forecast tonnes), the raw feedstock equation relies on a fixed industrial consumption intensity of 32.0 MMBtu per tonne. At our baseline AECO gas assumption of illustrative AECO gas assumption, the raw energy cash requirement sits at just \$38.40 per tonne. When matched against the sticky \$141.50 per tonne non-gas chemical conversion component, raw gas represents a minor 22.4% of direct asset-level supplier costs.

Mathematically, a localized 10% inflation shock to the AECO gas benchmark (moving from \$1.20 to \$1.32/MMBtu) expands total corporate COGS by a negligible 0.25% (an absolute dollar shift from \$17.818 billion to \$17.865 billion). Conversely, a parallel 10% expansion in non-gas processing inputs expands corporate COGS by a far more substantial 1.01%.

Competitive Positioning & Cost Advantages

To isolate the operational dynamics driving North American fertilizer equities, we evaluate the physical footprint of Nutrien Ltd. and CF Industries Holdings, Inc. While Nutrien manages a highly diversified agronomic portfolio across multi-nutrient extraction (Potash/Phosphate) and global retail networks, CF Industries operates as a highly consolidated, pure-play nitrogen manufacturing entity. This foundational divergence in corporate structure dictates their respective capital allocations, asset efficiencies, and margin behaviors across commodity cycles.

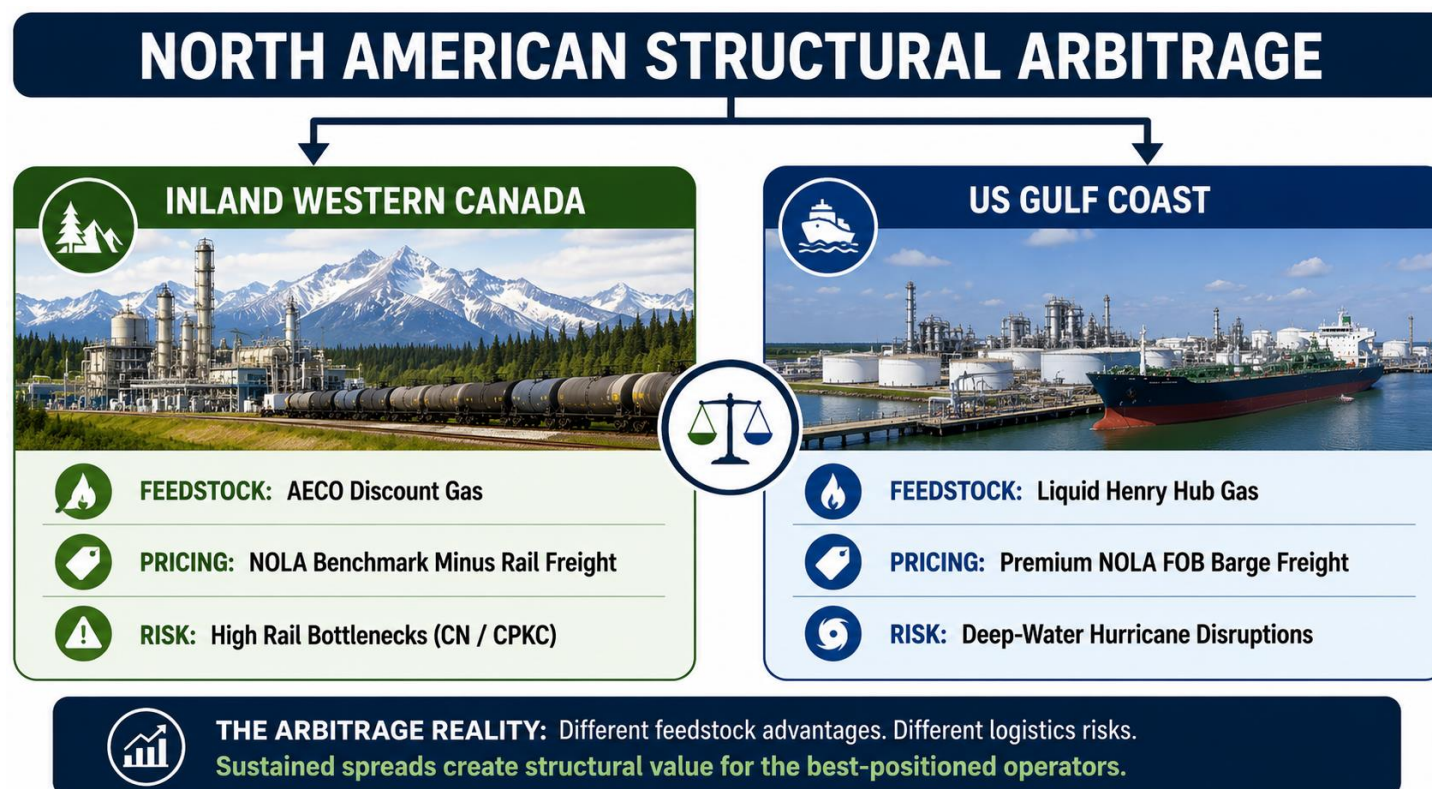
The table below outlines the structural asset allocations, nameplate processing capacities, and regional jurisdictions of both producers:

Producer	Core Production Facilities	Geographic Location	Primary Products	Nameplate Upstream Capacity (Annual)	Operational Baseline Utilization Rate	Logistic / Jurisdictional Framework
Nutrien (NTR)	Redwater Complex, Carseland Hub, Joffre Upgrader, Trinidad Fleet	Alberta, Canada & Point Lisas, Trinidad	Ammonia, Urea, UAN, AN, Diesel Exhaust Fluid (DEF)	7.30 MM Tonnes (G.Ammonia Baseline)	89.0% – 93.0%	AECO & Henry Hub Feedstock ; Direct integration into 2,000+ proprietary global retail storefronts.
CF Industries (CF)	Medicine Hat Complex, Donaldsonville Mega-Hub, Port Neal, Verdigris	Alberta, Canada & US Gulf Coast/Midwest	Ammonia,Urea, UAN, Nitric Acid	10.50 Million Short Tons (~9.52MM Tonnes)	95.0% – 97.0%	100% Henry Hub Asset Baseline (Excl. Med Hat Hub) ; Wholesale distribution via pipeline, river barge, and Class-1 rail channels.

NTR vs. CF INDUSTRIES – COMPETITIVE MATRIX		
 OPERATIONAL CATEGORY	<i>Nutrien</i> (NTR)	 CF INDUSTRIES (CF)
 Core Moat & Identity	 Vertical Integration	 Pure-Play Nitrogen Scale
 Feedstock Base	 AECO / WCSB Blended Hubs	 Henry Hub Dominated
 Nitrogen Capacity	 ~7.3 Metric Tonnes	 ~10.5 Short Tons
 Retail Network Assets	 2,000+ Captive Storefronts	 None (Wholesale Only)
 Distribution Model	 Direct-to-Grower Network	 Bulk Wholesale Logist.
 Potash / Phos Exposure	 Yes (Multi-Nutrient Base)	 No (100% Nitrogen Focus)
 Carbon Sequestration	 ACTL System Integrated	 Blue Ammonia Retrofits

Logistics & Supply Chain Considerations

The core operational edge for Western Canadian assets belongs to a systemic geographic advantage. Both Nutrien's Redwater/Carseland facilities and CF Industries' Medicine Hat complex sit directly adjacent to the landlocked AECO natural gas pricing hub. Because Western Canadian Sedimentary Basin (WCSB) production is structurally restricted by pipeline export capacity relative to the highly liquid US Gulf Coast (Henry Hub), local natural gas trades at a structural basis discount.



- The Feedstock Advantage: While global nitrogen costs are set by volatile European TTF or imported LNG, local operations capture a permanent feedstock cushion by drawing discounted AECO pricing assumptions AECO gas. Nutrien leverages its regional orientation by anchoring its Nitrogen segment COGS on 70% AECO Gas / 30% Henry Hub Gas. Conversely, CF Industries' consolidated cost stack operates on 100% Henry Hub exposure for its US production fleet, while its isolated Medicine Hat asset absorbs pure local AECO pricing. Both firms maintain a fixed chemical processing intensity of 32.0 MMBtu per tonne of ammonia synthesized, isolating raw gas price differentials as the fundamental driver of margin outperformance.
- The NOLA Netback & Valuation Modeling Alignment: North American nitrogen products are priced off the New Orleans (NOLA) FOB barge benchmark. Inland Canadian producers must sell their products at a localized freight discount relative to NOLA to cover the cost of long-haul rail transportation to key farming belts. However, because their local natural gas is so deeply discounted, the feedstock savings consistently outpace the rail freight penalties. This dynamic ensures that Western Canadian manufacturing plants maintain some of the widest cash margins on the global cost curve. To align modeling outputs with international macro indicators, the price baseline utilized across both corporate sheets represents the FOB NOLA equivalent price clearing in the domestic market, which is driven by global cost-curves linked back to European TTF margins. The pricing framework presented is illustrative and intended for scenario analysis rather than precise forecasting.

Jurisdictional Tax Risk & Carbon Exposure

The pure-play framework of CF Industries alters its risk profile relative to Nutrien when exposed to regional regulatory changes. In Canada, heavy industrial emitters face an output-based carbon pricing system. While Nutrien can dilute its regulatory compliance overhead across its global retail and multi-nutrient extraction operations (utilizing its **24.7% effective tax rate** framework), CF Industries' Canadian footprint is concentrated heavily within its carbon-intensive Medicine Hat chemical synthesis works under a leaner corporate **19.7% effective tax rate**.

To manage this risk, CF Industries is deploying targeted capital toward carbon capture and underground storage (CCUS) projects, laying the groundwork for a transition to low-carbon "blue" ammonia. This capital allocation protects their asset valuations from heavy regulatory compliance penalties while positioning the firm to capture premium prices in emerging global industrial clean energy and marine fuel export markets.

Logistics Vulnerabilities: Rail Bottleneck

Operating asset networks in Western Canada introduces severe logistical constraints. Unlike US Gulf Coast peers that enjoy direct access to flexible deepwater shipping and low-cost Mississippi River barge networks, Canadian infrastructure is almost entirely landlocked. Consequently, both Nutrien and CF Industries depend heavily on a dual-carrier railroad framework managed by Canadian National Railway (CN) and CPKC Railway.

While freight costs are tracked as a linear variable cost in financial models, real-world logistics operate on step-function constraints. Because Western Canada depends on a two-carrier railroad layout (CN and CPKC), a winter blizzard or rail strike acts as a hard capacity cap. If physical rail cars stop moving for more than 10 to 14 days, the factories cannot ship their products. Because storage domes have physical volume limits, the plants are forced to shut down production temporarily—even if their discounted AECO gas costs are highly favorable. This translates a logistics bottleneck into a temporary drop in capacity utilization rates.



Cost Structure Analysis

1. Reconciling Cost Distribution to Consolidated Outflows

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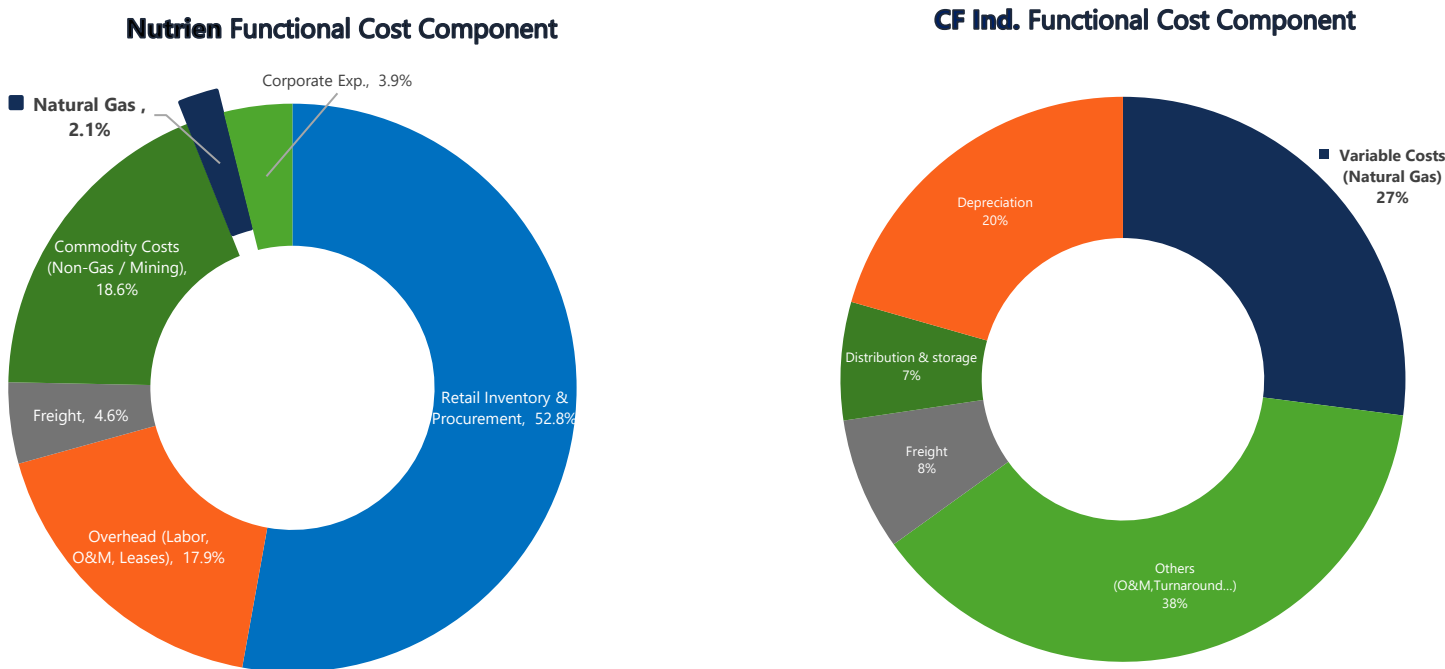
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2. Upstream Unit Economics & Consolidated Cost Anatomy

A granular analysis of the 2026F Forecast Cost Schedule reveals a distinct operational paradox within Nutrien's financial architecture: Consolidated corporate margins are highly insulated from domestic natural gas price spikes, yet remain completely unanchored on the global margin expansion curve.

For the 2026 forecast period, Nutrien's total consolidated Cost of Goods Sold (COGS) is projected at \$17.64 billion. A structural breakdown of this outflow shows that downstream asset classes—specifically the Downstream Retail Segment COGS at \$13.05 billion (occupying 74% of unadjusted cash costs) and Upstream Fixed Overhead at \$4.41 billion—overwhelmingly dominate corporate spending.

Within the upstream Nitrogen production stack, the raw feedstock equation relies on a fixed industrial consumption intensity of 32.0 MMBtu per tonne. At a baseline AECO gas assumption of \$1.20/MMBtu, the raw energy cash requirement sits at just \$38.40 per tonne. When matched against sticky non-gas chemical conversion components, raw gas represents a minor share of direct asset-level supplier costs.



Upstream Unit Economics & Consolidated Cost Anatomy

A granular analysis of our 2026F Forecast Cost Schedule reveals a distinct operational paradox within Nutrien's financial architecture: Consolidated corporate margins are highly insulated from domestic natural gas price spikes, yet remain completely unanchored on the global margin expansion curve.

For the 2026 forecast period, Nutrien's total consolidated Cost of Goods Sold (COGS) is projected at \$17.64 billion. A structural breakdown of this outflow shows that downstream asset classes—specifically the Downstream Retail Segment COGS at \$13.05 billion (occupying 74% of unadjusted cash costs) and Upstream Fixed Overhead at \$4.41 billion—overwhelmingly dominate corporate spending.

Within the upstream Nitrogen production stack (\$11.01 million forecast tonnes), the raw feedstock equation relies on a fixed industrial consumption intensity of 32.0 MMBtu per tonne. At our baseline AECO gas assumption of illustrative AECO gas assumption, the raw energy cash requirement sits at just \$38.40 per tonne. When matched against the sticky \$141.50 per tonne non-gas chemical conversion component, raw gas represents a minor 2.1% of direct asset-level supplier costs.

Mathematically, a localized 10% inflation shock to the AECO gas benchmark (moving from \$1.20 to \$1.32/MMBtu) expands total corporate COGS by a negligible 0.25% (an absolute dollar shift from \$17.638 billion to \$17.680 billion). Conversely, a parallel 10% expansion in non-gas processing inputs expands corporate COGS by a far more substantial 1.01%.

Reconciling Freight & Distribution Cost Items

To trace the quantitative cash impact of the logistical vulnerabilities discussed above, the forecast models track specific, divergent freight cost mechanisms across the 2025F–2026F period:

▣ Nutrien (NTR) Freight Schedule:

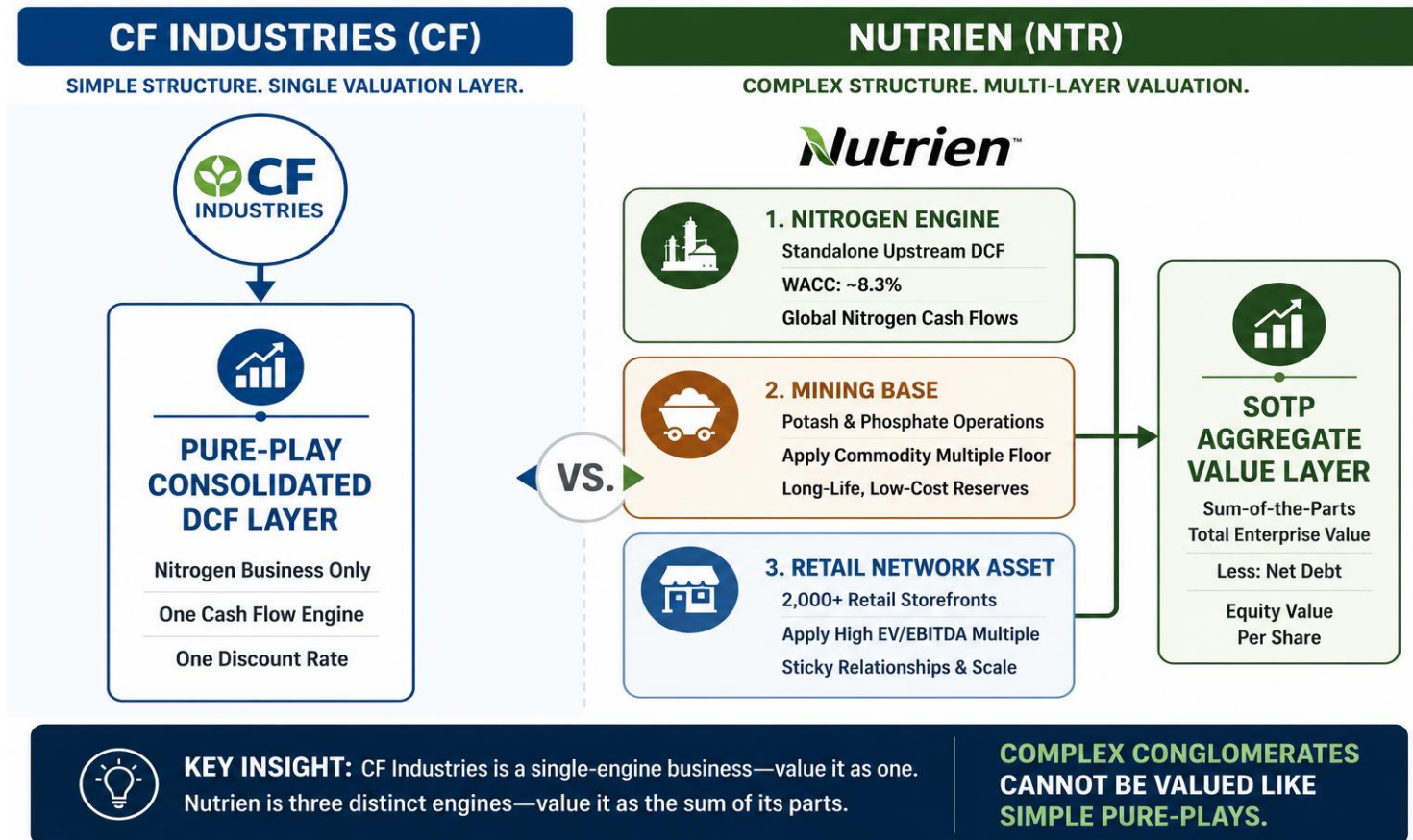
The standalone Nitrogen segment incurs an isolated freight penalty of **\$514M in 2025F**, which increases to **\$541M in 2026F** (modeled freight expense), reflecting changing internal cross-border rail paths and logistics optimization. On a consolidated holding-company basis, Nutrien's total corporate freight burden reaches **\$1,097M in 2025F and \$1,140M in 2026F**, driven heavily by the bulk movement of Potash out of Saskatchewan to West Coast ocean terminals.

▣ CF Industries (CF) Freight & Distribution Schedule:

Operating as a concentrated, high-volume wholesale manufacturer, CF maintains an ultra-lean shipping profile, booking direct outbound freight costs of **\$327M in 2025F and \$334M in 2026F** (modeled freight expense). However, because CF does not possess a captive retail store network to absorb regional drop-offs, it must absorb distinct **Storage and Distribution outlays of \$295M in 2025F and \$297M in 2026F**. This reflects the cost of maintaining a massive network of leased inland rail tanks, river domes, and regional field warehouses to hold product ahead of sudden, weather-dependent seasonal application windows.



Operational Risk Assessment



1. SOTP vs. Corporate DCF Valuation Methodology

A simple, blended corporate Discounted Cash Flow (DCF) model fails completely for a diversified giant like Nutrien because it masks the true standalone value of its underlying business units. To prevent conglomerate discount erosion, the valuation layer must be broken into two distinct architectural approaches:

- **CF Industries (The Clean Corporate DCF):** Because CF Industries is a tightly focused, pure-play nitrogen manufacturer, its valuation is cleanly captured by a consolidated Unlevered Corporate DCF model. Operating cash flows are driven symmetrically by global urea/ammonia benchmark prices and Henry Hub gas inputs, discounting the entire enterprise cleanly under a uniform corporate 7.2% WACC.
- **Nutrien (The Mandatory Sum-of-the-Parts Architecture):** Nutrien requires a strict Sum-of-the-Parts (SOTP) model to isolate its structural asset classes. It isolates the Upstream Nitrogen Engine independently using an 8.3% WACC to reflect pure commodity price volatility. It then layers on distinct, flat-run valuation multiples floor for the Potash and Phosphate mining operations. Finally, it applies a premium, low-volatility EV/EBITDA multiple to the massive Nutrien Ag Solutions Retail Network to reflect its stable, recurring direct-to-grower cash flows.
- **The Valuation Reconciliation:** The architectural model stacks these three standalone enterprise values, adds the captive asset value of Nutrien Financial, and backs out total holding-company net debt to yield a true, uncompromised corporate target share price.

2. Technical Modeling Assumptions & Core Parameters

To maintain perfect mathematical alignment across the forward schedules, the underlying valuation engine implements the following explicit criteria:

- The Segment Valuation Divergence Framework: Because Nutrien's core Nitrogen asset cash flows are modeled down to a standalone 8.3% Nitrogen WACC, their enterprise asset performance must be separated from the consolidated parent company's 7.69% WACC to avoid artificial capital-cost inflation. CF Industries' enterprise cash generation is discounted cleanly at a corporate 7.2% WACC, capturing its focused, single-segment financial risk profile.
- Multi-Nutrient Modeling Adjustments (NTR Parent Tab): To preserve the mathematical integrity of the integrated three-statement model without over-building ancillary sheets, the multi-nutrient segments are locked to constant historical performance run-rates derived from audited parameters:
 - **Potash Cost Stack:** Anchored to AECO natural gas pricing with a constant, contractually fixed mining consumption intensity of 1.20 MMBtu per product tonne.
 - **Phosphate Cost Stack:** Driven by AECO natural gas inputs utilizing a standardized, verified acidulation consumption intensity of 0.80 MMBtu per product tonne.
- Valuation Multiples Baseline: In the final peer index tab, the model evaluates trading multiples on an asset-to-asset basis. This allows the model to contrast the pure chemical cash generation of CF Industries directly against Nutrien's multi-tier upstream asset blocks before final equity valuation outputs are processed through the forward financial schedules.

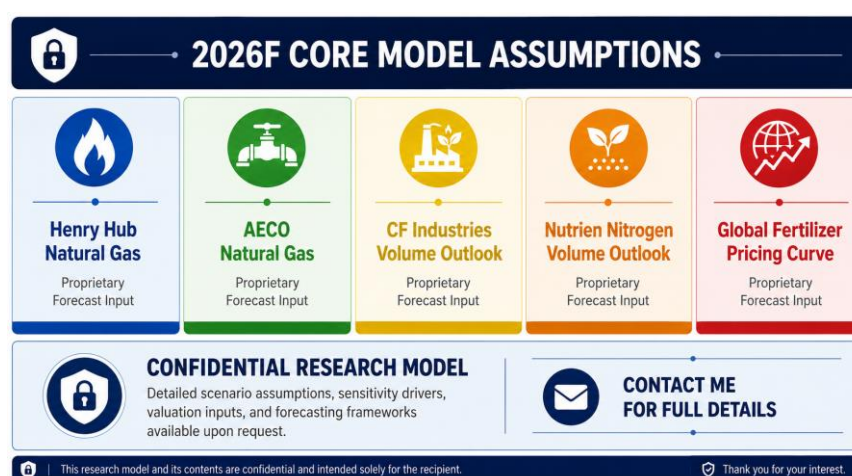
Financial Modeling & Scenario Analysis

Forecast Methodology

To transition our structural asset analysis into a dynamic financial forecasting framework, we translate geographic moats and logistical constraints into explicit modeling boundaries. Rather than anchoring the report to a single static forecast, the underlying model architecture initializes a flexible, multi-variable input layer. This allows for seamless real-time shifting across three distinct strategic paths (Base, Best, and Worst cases). Values shown are illustrative scenario inputs used for sensitivity analysis and do not represent price forecasts.

3.2 Key Modeling Assumptions

- The table below outlines the high-level operational levers and qualitative guidelines driving the model's 2026F directional scenarios:



METRIC	NTR STANDALONE NITROGEN 2026					CF INDUSTRIES 2026				
	Base	Best	% CHANGE TO BASE	Worse	% CHANGE TO BASE	Base	Best	% CHANGE TO BASE	Worse	% CHANGE TO BASE
GROSS PROFIT MARGIN	35.4%	41.8%	18.3%	29.3%	(17.2%)	42.8%	52.2%	22.0%	34.9%	(18.5%)
EBITDA	2,205	2,569	16.5%	1,870	(15.2%)	3,792	4,642	22.4%	3,097	(18.3%)
EBITDA MARGIN	44.4%	50.6%	14.1%	38.5%	(13.2%)	49.4%	58.9%	19.2%	41.4%	(16.1%)
NET INCOME MARGIN	21.0%	26.5%	26.3%	13.8%	(34.1%)	25.7%	33.4%	30.1%	19.2%	(25.3%)
UNLEVERED FCF	1,163	1,472	26.5%	876	(24.7%)	1,394	2,926	109.9%	272	(80.5%)
GROSS MARGIN PER TONNE	402	402	(0.1%)	403	0.2%	403	402	(0.0%)	402	(0.0%)
MANUFACTURING EFFECIANCY	1.55	1.72	11.0%	1.42	(8.4%)	1.75	2.10	20.0%	1.54	(12.0%)
EV / EBITDA	6.56	8.68	32.3%	1.26	(80.8%)	6.64	8.88	33.9%	2.43	(63.4%)
WC / REV	7.7%	8.1%	4.6%	7.4%	(4.3%)	3.1%	3.7%	20.9%	2.5%	(17.2%)



Key Takeaway: CF Industries delivers stronger upside scenarios and greater cash flow leverage, while NTR maintains balanced resilience across cycles with steadier downside protection and stable operational efficiency.

Sensitivity Analysis

To stress-test the structural earnings power of both asset networks under extreme macroeconomic conditions, we deploy a series of two-dimensional Excel data matrices. These tables cross-reference primary feedstock cost drivers against global market clearing realizations to isolate the absolute downside floors and upside limits of both corporate entities.

Core Energy Feedstock vs. EBITDA Margin Sensitivity

Methane represents approximately 70% to 80% of pure chemical cash COGS for nitrogen synthesis. Shifting regional gas benchmarks serve as the primary driver of margin expansion or contraction.

The analytical tables below track how simultaneous changes in Henry Hub natural gas pricing and NOLA Urea sales realizations impact the standalone operational EBITDA performance of Nutrien's Nitrogen Segment versus CF Industries' pure-play consolidated framework:

Nutrien: Nitrogen Segment EBITDA Margin Sensitivity

		Fertilizer Sales Price (\$/Tonne)				
		\$310	\$340	\$370	\$400	\$430
Henry Hub Gas Price (\$/MMBtu)	2.00	30.2%	36.3%	41.4%	45.7%	49.5%
	2.50	29.1%	35.3%	40.5%	44.9%	48.7%
	3.00	28.0%	34.3%	39.6%	44.1%	47.9%
	3.50	27.0%	33.4%	38.7%	43.3%	47.2%
	4.00	25.9%	32.4%	37.8%	42.4%	46.4%

CF Industries: EBITDA Margin Sensitivity to Henry Hub

		Fertilizer Sales Price (\$/Tonne)				
		\$310	\$340	\$370	\$400	\$430
Henry Hub Gas Price (\$/MMBtu)	2.00	41.0%	46.2%	50.6%	54.3%	57.5%
	2.50	37.6%	43.1%	47.7%	51.7%	55.0%
	3.00	34.3%	40.1%	44.9%	49.1%	52.6%
	3.50	30.9%	37.0%	42.1%	46.5%	50.2%
	4.00	27.6%	34.0%	39.3%	43.9%	47.8%

The comparative sensitivity framework highlights a fundamental distinction between the operating architectures of Nutrien's Nitrogen segment and CF Industries. While both businesses remain highly exposed to nitrogen pricing and natural gas feedstock costs, the composition of their underlying cost structures produces materially different earnings outcomes across commodity cycles.

➤ CF Industries: Pure-Play Nitrogen Operating Leverage

CF Industries represents one of the industry's highest-beta exposures to the North American nitrogen cycle. Its manufacturing network is predominantly indexed to Henry Hub natural gas, causing operating earnings to exhibit a highly linear relationship with domestic feedstock inflation. Under favorable fertilizer pricing conditions, this structure maximizes incremental margin capture and EBITDA expansion. However, during periods of compressed nitrogen realizations or rising gas costs, profitability deteriorates rapidly as feedstock inflation flows directly through the cost base. Consequently, CF relies heavily on capacity utilization, export volumes, and favorable nitrogen spreads to sustain earnings momentum.

➤ Nutrien Nitrogen: Diversified Feedstock Architecture

In contrast, Nutrien's Nitrogen segment benefits from a geographically diversified feedstock portfolio anchored by a blended sourcing structure of approximately 60% AECO-linked Western Canadian gas and 40% Henry Hub-linked U.S. gas.

This sourcing mix provides a structural cost advantage during periods of North American gas inflation because AECO frequently trades at a meaningful discount to Henry Hub due to regional transportation constraints and recurring basin bottlenecks. As a result, Nutrien's nitrogen platform maintains a lower marginal production cost and a more resilient EBITDA floor throughout cyclical downturns.

➤ **Margin Asymmetry and Downside Protection**

The sensitivity analysis demonstrates that Nutrien's primary advantage is not superior upside participation, but rather superior downside protection.

Under a constant nitrogen selling price assumption of approximately \$370 per tonne, increasing Henry Hub gas prices from \$2.00/MMBtu to \$4.00/MMBtu reduces CF Industries' operating margin from approximately 50.6% to 39.3%, representing a contraction of roughly 1,130 bps. Over the same cost shock, Nutrien's Nitrogen segment experiences a materially smaller margin decline from approximately 41.4% to 37.8%, a contraction of approximately 360 bps.

Under the assumptions employed the economic value of Nutrien's blended feedstock architecture. By maintaining substantial exposure to discounted Western Canadian gas, Nutrien absorbs a smaller proportion of U.S. Gulf Coast cost inflation than CF Industries, preserving profitability when nitrogen spreads compress.

➤ **Investment Interpretation**

From an equity perspective, the two companies provide distinct exposure profiles:

- CF Industries offers greater earnings sensitivity to widening nitrogen spreads and favorable commodity pricing environments making it the higher-beta vehicle for investors seeking maximum exposure to a commodity upcycle.
- Nutrien provides a more balanced risk-return profile, combining nitrogen exposure with a structurally advantaged feedstock position and reduced sensitivity to U.S. natural gas inflation.

Accordingly, CF Industries can be viewed as the superior vehicle for capturing cyclical upside, whereas Nutrien's nitrogen platform exhibits stronger defensive characteristics and a more durable earnings base during periods of commodity, energy, or margin compression. This distinction becomes increasingly valuable when global nitrogen markets transition from expansionary conditions toward mid-cycle or recessionary environments.

Urea-to-Gas Sensitivity: Net EBITDA Output

To translate percentage margin outcomes into tangible earnings power, the sensitivity framework converts operating assumptions into absolute EBITDA generation across varying fertilizer pricing and natural gas environments. This approach provides a clearer assessment of each company's cash-generation capacity throughout the nitrogen cycle and highlights the trade-off between downside resilience and upside earnings leverage.

Nutrien Nitrogen Segment EBITDA (\$ Millions)						CF Ind. EBITDA (\$ Millions)							
Henry Hub Gas Price (\$/MMBtu)	Fertilizer Sales Price (\$/Tonne)					Henry Hub Gas Price (\$/MMBtu)	Fertilizer Sales Price (\$/Tonne)						
	\$310	\$340	\$370	\$400	\$430		\$310	\$340	\$370	\$400	\$430		
	2.00	1,156	1,524	1,892	2,260		2,629	2.00	2,424	2,996	3,568	4,140	4,713
	2.50	1,115	1,483	1,851	2,219		2,588	2.50	2,225	2,798	3,370	3,942	4,515
	3.00	1,074	1,442	1,810	2,178		2,547	3.00	2,027	2,600	3,172	3,744	4,317
	3.50	1,033	1,401	1,769	2,137		2,506	3.50	1,829	2,402	2,974	3,546	4,118
	4.00	992	1,360	1,728	2,097		2,465	4.00	1,631	2,204	2,776	3,348	3,920

➤ Downside Floor Assessment

Under a stressed commodity environment characterized by \$310/tonne fertilizer realizations and \$3.50/MMBtu Henry Hub natural gas, Based on model assumptions, CF Industries generates approximately \$1.83 billion of annual EBITDA, while Nutrien's standalone Nitrogen segment produces approximately \$1.03 million. Although both businesses remain cash-generative under adverse market conditions, CF's substantially larger production base and nitrogen concentration provide a materially higher absolute EBITDA floor.

Importantly, these figures should not be interpreted as a direct corporate comparison. The Nutrien results reflect only the Nitrogen operating segment and exclude contributions from Potash, Retail, and Phosphate operations. As a result, consolidated Nutrien earnings would be materially higher than the standalone nitrogen sensitivity suggests.

➤ Upside Earnings Capture

The same framework illustrates the differing upside profiles of the two operating models. Under a constructive pricing environment of \$430/tonne fertilizer realizations and \$2.00/MMBtu Henry Hub gas, Nutrien's Nitrogen segment generates approximately \$2.63 billion of EBITDA, compared with approximately \$4.71 billion for CF Industries.

This outcome highlights CF's superior earnings torque to favorable nitrogen spreads. As fertilizer prices rise and feedstock costs remain contained, the company's concentrated nitrogen exposure converts incremental revenue into EBITDA at an accelerated rate. Consequently, CF provides greater participation in periods of tightening global nitrogen balances and elevated agricultural profitability.

➤ Corporate-Level Interpretation

The sensitivity analysis ultimately highlights a distinction between earnings concentration and earnings diversification.

- CF Industries derives the majority of its value from a large-scale nitrogen manufacturing platform, making absolute EBITDA generation highly responsive to changes in nitrogen spreads. This structure maximizes upside participation during favorable commodity cycles but also concentrates corporate performance within a single product category.

- Nutrien's standalone Nitrogen segment generates lower absolute EBITDA across most modeled scenarios; however, the segment operates within a broader vertically integrated enterprise. Nutrien Ag Solutions, the company's retail distribution network, provides an additional earnings stream that is not directly captured in the nitrogen sensitivity matrix. During periods of weaker fertilizer pricing, retail operations can partially offset manufacturing pressure through distribution margins, customer financing activity, and recurring agronomic services revenue.

➤ **Investment Implications**

Viewed through a cash-generation lens, CF Industries offers greater absolute earnings leverage to nitrogen market strength, while Nutrien provides a more diversified earnings framework supported by downstream retail integration. Accordingly, CF represents the higher-conviction pure-play nitrogen exposure, whereas Nutrien's broader operating structure may provide greater earnings stability across varying commodity environments.

Margin & EBITDA Drivers

To stress-test the structural revenue defenses of both networks against international supply shocks, we analyze the cross-commodity interaction between European Title Transfer Facility (TTF) natural gas pricing, domestic New Orleans (NOLA) barge benchmarks, and total sales volumes.

Global Cost-Curve Parity Analysis:

TTF-Driven Margin Expansion EBITDA Margin

European ammonia and nitrogen producers operating on TTF-indexed natural gas are often viewed as marginal producers within the global nitrogen value chain. As a result, fluctuations in European gas prices materially influence international fertilizer clearing prices, establishing the economic floor from which global nitrogen pricing is derived.

The sensitivity framework demonstrates that both Nutrien and CF Industries benefit disproportionately when TTF natural gas prices rise relative to North American feedstock benchmarks. Because their production platforms are primarily supplied by discounted Henry Hub and AECO gas, increases in European energy costs widen the global feedstock arbitrage, allowing North American producers to capture higher nitrogen realizations without experiencing a commensurate increase in production costs.

Structural Margin Expansion Through Cost-Curve Dislocation

Nutrien Nitrogen Segment Global Cost-Curve Parity

		Henry Hub Gas Price				
		\$2.00	\$2.50	\$3.00	\$3.50	\$4.00
Global parity Gas Price (\$/MMBtu)	12.20	37.9%	36.9%	36.0%	35.0%	34.1%
	14.20	42.2%	41.4%	40.5%	39.6%	38.7%
	16.20	46.0%	45.2%	44.4%	43.5%	42.7%
	18.20	49.3%	48.6%	47.8%	47.0%	46.2%
	20.20	52.3%	51.5%	50.8%	50.1%	49.3%

CF Ind. Global Cost-Curve Parity

		Henry Hub Gas Price				
		\$2.00	\$2.50	\$3.00	\$3.50	\$4.00
Global parity Gas Price (\$/MMBtu)	12.20	47.6%	44.6%	41.7%	38.7%	35.7%
	14.20	51.3%	48.6%	45.8%	43.0%	40.3%
	16.20	54.5%	52.0%	49.4%	46.8%	44.2%
	18.20	57.4%	55.0%	52.5%	50.1%	47.7%
	20.20	59.9%	57.6%	55.3%	53.0%	50.8%

- The data highlights a powerful operating dynamic: as TTF gas increases from \$12.20/MMBtu to \$20.20/MMBtu, global nitrogen clearing prices move materially higher, lifting profitability across the North American nitrogen industry.
 - Nutrien's Nitrogen segment, EBITDA margins expand from approximately 34–38% under the lowest TTF environment to 49–52% under the highest TTF scenario, depending on prevailing Henry Hub prices. This demonstrates the segment's ability to convert international cost inflation into incremental earnings despite maintaining a comparatively defensive cost structure.
 - CF Industries exhibits an even stronger response. Under the same TTF escalation scenario, EBITDA margins increase from approximately 36–48% to 51–60%, reflecting the company's higher concentration in merchant nitrogen production and greater exposure to wholesale nitrogen pricing.

➤ **Margin Leverage and Competitive Positioning**

The comparative analysis reinforces the distinct economic roles of the two companies within the nitrogen industry.

- CF Industries functions as the industry's highest-conviction pure-play on global nitrogen spreads. Its concentrated asset base, large-scale ammonia footprint, and wholesale distribution model maximize earnings sensitivity to rising international fertilizer prices. When European producers move further up the global cost curve, CF captures an outsized share of the resulting margin expansion.
- Nutrien's Nitrogen segment, while also benefiting from elevated TTF pricing, delivers a more balanced risk profile. The combination of discounted AECO feedstock exposure, downstream integration, and diversified nutrient operations reduces earnings volatility while still allowing substantial participation in favorable global nitrogen pricing environments.

➤ **Cross-Asset Interpretation**

The interaction between Henry Hub and TTF pricing reveals two distinct sources of value creation: Domestic Feedstock Advantage (Henry Hub / AECO): Lower North American gas prices directly reduce production costs and protect operating margins.

International Cost-Curve Inflation (TTF): Higher European gas prices raise global nitrogen clearing prices and expand producer profitability.

The most favorable earnings environment for both companies occurs when European TTF prices remain elevated while North American gas prices remain relatively contained. Under this scenario, the spread between global nitrogen selling prices and North American production costs reaches its widest level, generating peak margin capture.

➤ **Investment Interpretation**

The matrix does not fully prove that TTF is more important than Henry Hub. While Henry Hub primarily influences the cost side of the equation, TTF influences the global marginal producer and therefore the industry's pricing structure.

➤ **Consequently:**

- CF Industries offers the highest leverage to a widening g
- Nutrien provides exposure to the same macro tailwind while maintaining a stronger margin floor through feedstock diversification and broader business integration.

Operating Leverage Assessment

To isolate the true fixed-cost absorption capacity of both operators, we execute a two-dimensional sensitivity check crossing total sales volumes against final blended fertilizer selling prices:

Nutrien Nitrogen Segment Leverage Profile						CF Ind. Leverage Profile							
Fertilizer Sales Price (\$/Tonne)						Fertilizer Sales Price (\$/Tonne)							
Total Nitrogen Sales Volume (\$MMBtu)		\$310	\$340	\$370	\$400	\$430	Total Nitrogen Sales Volume (\$MMBtu)		\$310	\$340	\$370	\$400	\$430
	11.00	19.2%	26.3%	32.2%	37.3%	41.6%		18.30	31.8%	37.8%	42.8%	47.1%	50.8%
	11.50	22.7%	29.5%	35.2%	40.0%	44.1%		18.70	33.1%	39.0%	43.9%	48.1%	51.8%
	12.00	25.9%	32.4%	37.8%	42.4%	46.4%		19.10	34.4%	40.2%	45.0%	49.1%	52.7%
	12.50	28.9%	35.1%	40.3%	44.7%	48.5%		19.50	35.6%	41.3%	46.0%	50.1%	53.6%
	13.00	31.6%	37.6%	42.6%	46.8%	50.5%		19.90	36.7%	42.3%	47.0%	51.0%	54.4%

The volume sensitivity framework isolates the earnings impact of production throughput and sales realizations by evaluating EBITDA margin outcomes across varying nitrogen sales volumes and fertilizer pricing environments. The analysis highlights a clear divergence in operating leverage characteristics between CF Industries' pure-play manufacturing model and Nutrien's vertically integrated nitrogen platform.

CF Industries: High-Beta Manufacturing Leverage

CF Industries exhibits a pronounced operating leverage profile driven by the scale and concentration of its nitrogen manufacturing network. Because a substantial portion of the company's cost structure is fixed, incremental volume growth generates meaningful margin expansion as fixed production costs are spread across a larger sales base.

Across the sensitivity matrix, EBITDA margins consistently expand as nitrogen throughput increases, reflecting the strong earnings conversion at a baseline price of \$370/Tonne, expanding total sales volumes from 18.30 to 19.90 million tonnes drives EBITDA margins from **42.8% to 47.0%** (420 bps expansion due purely to fixed-cost dilution) associated with higher asset utilization. This characteristic positions CF as a highly effective vehicle for capturing upside during periods of robust agricultural demand, favorable nitrogen pricing, and elevated export activity.

However, the same operating structure increases exposure to volume-related disruptions. Rail congestion, export terminal constraints, weather-related logistics interruptions, or unexpected plant outages can materially impair fixed-cost absorption, causing profitability to deteriorate more rapidly than at more diversified peers. Consequently, CF's earnings profile remains highly dependent on maintaining elevated operating rates and uninterrupted product movement throughout its distribution network.

Nutrien Nitrogen Segment: Asset Scale vs. Structural Integration

Paradoxically, within this specific model layout, Nutrien's Nitrogen segment displays higher mathematical volume operating leverage than CF Industries due to its smaller standalone volume base. At a constant pricing threshold of \$370/Tonne, scaling volume from top to bottom (**an ~18% increase in output**) expands Nutrien's EBITDA margins by 1,040 bps (from 32.2% to 42.6%). Meanwhile, a similar volume scale-up at CF Industries only yields a 420 bps margin expansion (from 42.8% to 47.0%), demonstrating how incremental tonnage additions are mathematically more powerful for fixed-cost dilution on Nutrien's less-scaled footprint.

However, in a live market environment, Nutrien's true underlying earnings sensitivity is structurally moderated by its vertically integrated operating model. Because a meaningful portion of upstream nitrogen production is distributed directly through the company's captive retail platform, Nutrien enjoys a secure internal demand channel. This captive downstream commercialization partially insulates the manufacturing operations from wholesale spot shocks, smoothing out the extreme margin volatility typically observed in pure-play commodity frameworks.

Cross-Asset Comparative Assessment

The volume sensitivity analysis reinforces the fundamental distinction between **operating leverage** and **earnings durability** across the two nitrogen producers. While both companies benefit from higher production throughput and stronger fertilizer pricing, the magnitude and stability of their earnings responses differ materially because of their underlying business models.

- **CF Industries** exhibits superior operating leverage, with EBITDA margins responding with higher absolute price torque (scaling from 31.8% to 50.8% across the \$310 to \$430 price band at baseline volume) rather than volume torque. Its concentrated manufacturing footprint enables exceptional fixed-cost absorption during periods of high asset utilization, creating significant earnings acceleration as industry conditions improve.
- **Nutrien's Nitrogen segment** preserves a higher degree of standalone operating volume sensitivity (+1,040 bps margin expansion across its volume range) in exchange for greater earnings stability. The company's vertically integrated production and retail network provides an internal demand channel that moderates the impact of wholesale volume fluctuations, resulting in a more resilient profitability profile across varying market environments.

➤ Investment Implication:

The sensitivity matrices suggest that CF Industries offers greater upside participation during expansionary phases of the nitrogen cycle, whereas Nutrien provides a more defensive earnings profile supported by diversified end-market exposure and enhanced volume visibility.

From a portfolio construction perspective, CF Industries should therefore be viewed as the **higher-beta** nitrogen investment, with financial performance more directly linked to production throughput, export demand, and industry utilization rates. Nutrien, by contrast, benefits from a partially insulated operating model in which downstream integration dampens volume-related earnings volatility and supports more consistent cash flow generation throughout the commodity cycle.

Accordingly, CF Industries offers greater upside torque when nitrogen fundamentals strengthen, while Nutrien's integrated platform provides stronger downside protection through superior demand visibility, internal product absorption, and reduced reliance on spot-market sales.

Strategic Interpretation

The operating leverage analysis ultimately reflects two distinct corporate strategies for creating shareholder value.

- **CF Industries** represents an industrial manufacturing optimization model, where value creation is driven by maximizing plant utilization, expanding nitrogen spreads, and converting incremental production into disproportionately higher earnings and free cash flow. This operating structure produces substantial upside during favorable pricing environments but also increases exposure to operational disruptions, logistics constraints, and cyclical demand weakness.
 - **Nutrien's Nitrogen segment** operates within a vertically integrated agricultural value chain, where manufacturing assets are supported by downstream retail distribution and long-term customer relationships. Although this integration moderates peak-cycle earnings acceleration relative to a pure-play producer, it improves throughput visibility, stabilizes fixed-cost absorption, and reduces earnings volatility across changing market conditions.
- **Ultimately**, the comparison highlights a clear strategic trade-off: CF Industries prioritizes operating efficiency and earnings torque, while Nutrien emphasizes earnings durability and cash-flow resilience. The choice between the two therefore depends less on absolute nitrogen market fundamentals than on an investor's preference for higher cyclical upside versus more consistent through-the-cycle performance.

Valuation & Strategic Insights

Relative Valuation Analysis

To translate the operating analysis into an institutional valuation framework, we benchmark forward-looking valuation multiples directly against capital efficiency and cash-generation metrics.

Valuation & Efficiency Multiples (2026F)	Nutrien Nitrogen (Standalone Segment)	Nutrien Parent (Consolidated Total)	CF Industries (Consolidated Total)
Market Value (Equity Blended \$M)	0	32,980	17,228
Enterprise Value (\$M)	12,299	44,227	20,399
Current/Model Price (\$)	-	65	109
EV / EBITDA	6.6	7.6	6.6
EV / EBIT	7.3	12.2	7.1
Unlevered Valuation (EV / FCFF)		11.1	9.0
Price / Free Cash Flow (P/FCF)	-	8.6	8.1
Price / Earnings (P/E)	-	14.0	8.7
Price / Sales (P/S)	-	1.2	2.2
Gross Margin Per Tonne (\$/t)	402	923	403
Manufacturing Efficiency Ratio	1.55	1.55	1.75
Working Capital / Revenue (%)	7.7%	15.7%	3.1%
Cash Conversion Cycle (Days)	21	98.3	-1.1
Return on Assets (ROA %)	7.7%	4.2%	13.9%
Return on Equity (ROE %)	-	8.6%	25.0%
WACC (%)	8.3%	7.69%	7.20%
Target Capital Structure (Debt/Eq)		%35-%65	%30-%70

The matrix compares three perspectives:

- Nutrien Nitrogen as a standalone upstream nitrogen platform.
- Nutrien Parent as the fully integrated potash, phosphate, nitrogen, and retail enterprise.
- CF Industries as a pure-play nitrogen producer with minimal downstream complexity.

This side-by-side framework allows investors to separate the economics of Nutrien's low-cost nitrogen franchise from the working-capital characteristics of its agricultural retail network.

Metrics are derived from model assumptions, public filings, consensus estimates, and illustrative scenario inputs. They should not be interpreted as investment recommendations.

Operating Efficiency Comparison

- Return Metrics: Consolidated Distortion vs. Segment Reality

At the consolidated level, CF Industries screens as the clear efficiency leader, generating approximately 13.9% ROA and 25.0% ROE, versus Nutrien Parent at roughly 4.2% ROA and 8.6% ROE.

However, this comparison masks the underlying economics of Nutrien's nitrogen business. When the nitrogen segment is isolated, it produces a competitive 7.7% ROA, materially above the consolidated corporate average. The gap is largely attributable to Nutrien Ag Solutions, which operates a seasonal distribution model requiring substantial inventory and receivables investment.

Key balance-sheet contrasts include:

Metric	Nutrien Parent	CF Industries
Working Capital / Revenue (%)	15.7%	3.1%
Cash Conversion Cycle (Days)	98.3	-1.1
Business Model	Integrated retail + nutrient distribution	Pure wholesale manufacturing

CF's negative cash conversion cycle reflects advance customer payments and forward supply contracts, allowing operations to be funded with minimal net working capital. Nutrien's integrated retail model requires significantly higher inventory and receivable balances, resulting in working capital equal to 15.7% of revenue versus only 3.1% for CF Industries. While this reduces reported capital efficiency, it also provides a broader earnings base and greater resilience across agricultural cycles.

The Multi-Nutrient Multiple Arbitrage

Isolating Nutrien's upstream nitrogen business reveals a valuation discrepancy that is obscured within the consolidated entity.

Valuation Metric	Multiple
Nutrien Nitrogen Standalone EV/EBITDA	6.6x
CF Industries Consolidated EV/EBITDA	6.6x

The market awards CF a premium for its simplified corporate structure, where manufacturing margins flow directly to shareholders without retail inventory drag.

In contrast, Nutrien's nitrogen assets possessing a structural feedstock advantage. A significant portion of Nutrien's production is supplied from Western Canadian AECO-linked natural gas, which historically prices below major international hubs. This creates a lower-cost position on the global nitrogen curve and provides meaningful downside protection during fertilizer downturns.

Cash Conversion and Capital Allocation

The gap between P/FCF (Nutrien: 8.6x; CF: 8.1x) reflects two fundamentally different capital-allocation models.

➤ CF Industries -- Pure-Play Cash Return Engine

CF's high manufacturing efficiency ratio (1.75x) converts a large share of gross profit into free cash flow. With limited need for retail infrastructure investment, excess cash is directed toward:

- Share repurchases
- Debt reduction
- Balance-sheet optimization

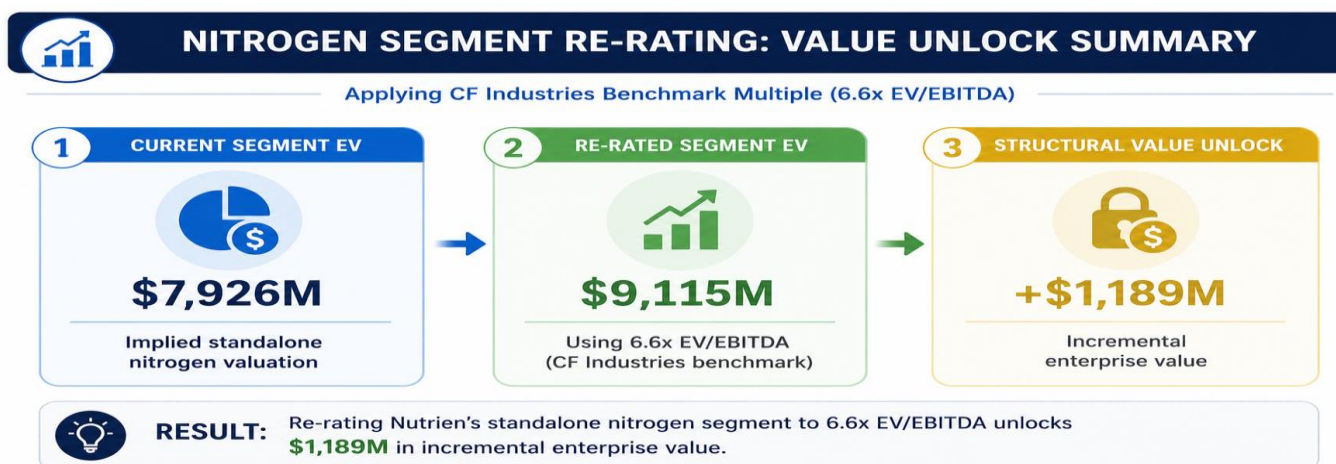
➤ Nutrien -- Integrated Cash-Flow Buffer

Nutrien's lower 11.1x EV/FCFF profile reflects the market's view of a more capital-intensive enterprise. However, the retail platform provides counter-cyclical earnings support when fertilizer prices weaken, partially stabilizing consolidated cash flow through agricultural input distribution and service revenues.

Sum-of-the-Parts Valuation

➤ Nutrien: Nitrogen Re-Rating Potential

Applying CF Industries' benchmark 6.6x EV/EBITDA multiple to Nutrien's standalone nitrogen segment suggests potential valuation upside under the model assumptions employed.



Using separate peer multiples for Retail and Potash/Phosphate operations produces a revised consolidated enterprise value of approximately \$44.2B.

After deducting net debt and dividing by the current share count, the SOTP framework yields a target value of approximately \$69/share, implying ~6% upside from the current model price of \$65.

➤ **CF Industries: Valuation Near the Upper Bound**

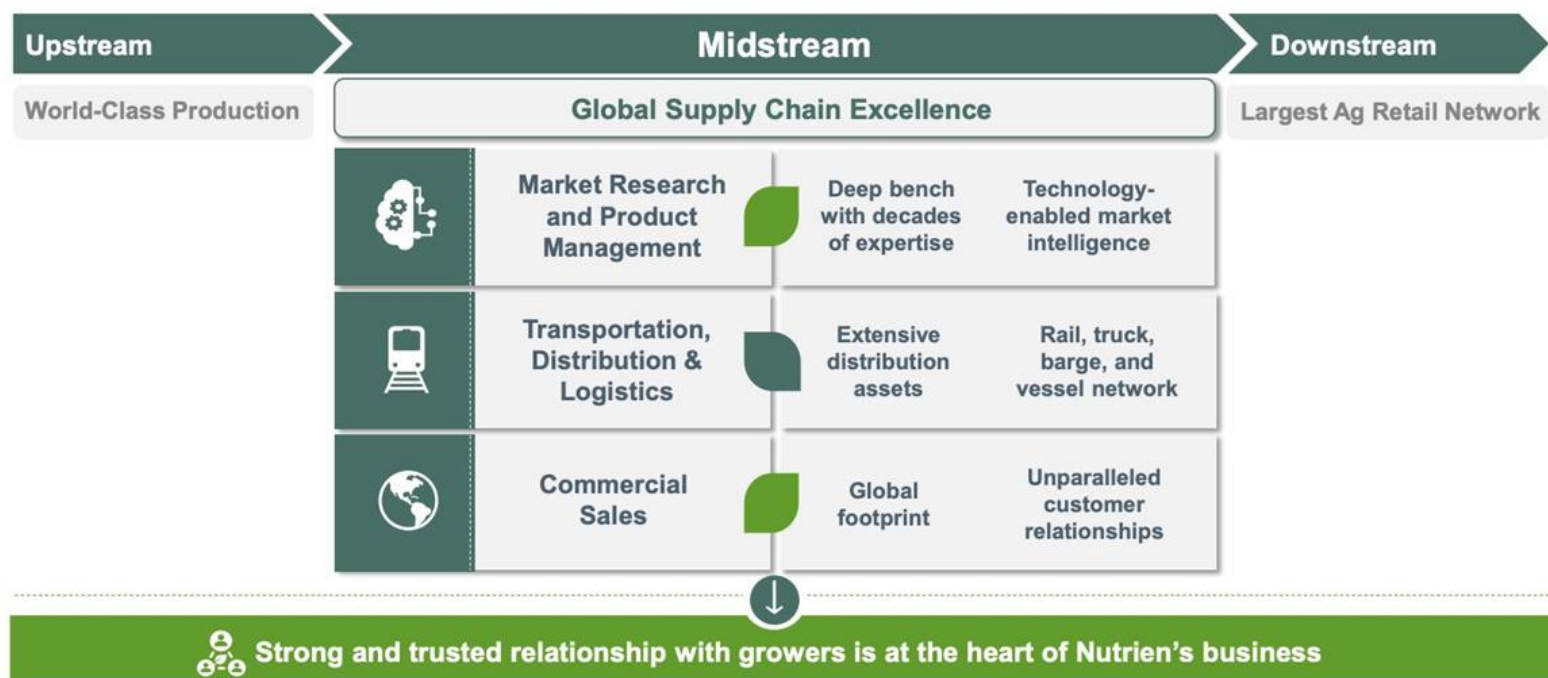
CF Industries already trades at a premium through-the-cycle valuation, reflecting investor recognition of its superior cash conversion and shareholder return profile.

Valuation Metric	Value
EV / EBITDA	6.6
Price / Free Cash Flow (P/FCF)	8.1
Modeling Price (\$)	109

At current levels, future upside is likely to be driven primarily by incremental capacity additions, nitrogen pricing, or sustained margin expansion rather than multiple re-rating.

Capital Allocation Assessment

The following framework summarizes how operating efficiency translates into capital allocation and, ultimately, valuation persistence.



➤ CF Industries: Volumetric Cash Return Focus

CF's negative cash conversion cycle and high manufacturing efficiency support a lean capital model. Management's priorities are expected to remain:

- Limit large-scale expansionary capex.
- Continue aggressive share repurchases.
- Maintain a conservative balance sheet (~30% debt / 70% equity).
- Preserve a low cost of capital through industry downturns.

➤ Nutrien: Working Capital Optimization Focus

Nutrien's strategic emphasis is less about adding nitrogen capacity and more about improving capital efficiency across the integrated platform.

Key priorities include:

- Reducing the 16.2% working-capital-to-revenue burden through supply-chain and inventory optimization.
- Using cash flow from the nitrogen segment to reduce leverage.
- Maintaining a balanced 35% debt / 65% equity capital structure.

Supporting a sustainable dividend while retail inventory cycles normalize.

Conclusion

Key Analytical Findings

Based on the assumptions used throughout this analysis, the model suggests a valuation disconnect between Nutrien's consolidated trading multiple and the implied valuation of its low-cost nitrogen assets.

While Nutrien Parent trades around 7.6x EV/EBITDA, the embedded nitrogen business is valued at only ~6.6x, reflecting its cost-advantaged feedstock exposure on the global nitrogen cost curve through AECO-linked feedstock access.

CF Industries remains a best-in-class cash generator trading at a fair premium for its operational simplicity and shareholder return profile. However, its current valuation already reflects much of that advantage.

For Nutrien, our SOTP analysis produces a \$69/share target value, representing approximately 6% upside from the current model price. The re-rating thesis is supported by:

- Low-cost Canadian nitrogen exposure.
- Defensive through-the-cycle cash margins.
- Ongoing working-capital optimization within the retail segment.
- A capital allocation roadmap focused on deleveraging and shareholder returns.

In our view, the market is appropriately valuing CF Industries' efficiency, but continues to underappreciate the intrinsic value of Nutrien's nitrogen franchise within the broader conglomerate structure.

Appendices

References & Data Sources :

- Bloomberg Intelligence: Green Markets Index Data (BI AGCH Hub).
- World Bank Group: Commodity Markets Outlook & Fertilizer Market Reports.
- Nutrien Ltd.: Corporate Reports, Annual Information Forms (Form 40-F), Sustainability Registries, and Investor Relations Guidance Schedules.
- CF Industries Holdings, Inc.: SEC Filings, Financial Statements, and Statutory Disclosures.
- U.S. Energy Information Administration (EIA): Short-Term Energy Outlook (STEO) Natural Gas Data.
- Government of Alberta: Economic Dashboard (Natural Gas Price Data).
- Auspice Capital: Canadian Natural Gas Index (Auspice NG Index).
- International Fertilizer Association (IFA): IFASTAT Global Fertilizer Market Statistics.